

FROBENIUS MULTIPLICITIES AND F -SIGNATURES OF GRASSMANNIANS OF TWO-PLANES

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ABSTRACT. Raedschelders, Špenko and Van den Bergh proved that the Plücker coordinate ring C_n of the Grassmannian $\mathrm{Gr}(2, n)$ has finite Frobenius representation type when the characteristic p satisfies $p \geq \max\{n - 2, 3\}$, and they determined the indecomposable summands of its Frobenius pushforwards up to nonzero multiplicity. We give nonrecursive finite formulas for every entry of the Frobenius multiplicity matrix and prove that each entry is a polynomial in p of degree at most $2n - 3$. Consequently the F -signature of C_n , and the asymptotic proportion of every indecomposable summand, is a single rational function of p throughout the characteristic range. Its limit as $p \rightarrow \infty$ is the leading coefficient of the first free multiplicity, which we evaluate in closed form through Bernoulli numbers. For $\mathrm{Gr}(2, 4)$ we find $s_p(C_4) = (13p^2 + 8)/(5(3p^2 + 2))$, and we calculate the analogous rational functions for $\mathrm{Gr}(2, 5)$ and $\mathrm{Gr}(2, 6)$ with computer assistance. We also construct the full characteristic polynomial from the Riemann–Roch weights and two residual matrices whose entries have degree at most $n - 1$ in p . In the sequel, we use this work to show rank-two determinantal rings have FFRT, and to calculate their F -signatures.

1. INTRODUCTION

Let k be an algebraically closed field of characteristic $p > 0$ and R a positively graded k -algebra with $R_0 = k$. Write $F_*^e R$ for R regarded as a module over itself through the e th Frobenius map and let $a_e(R)$ denote the maximal rank of an R -free summand of $F_*^e R$. The ring R has *finite Frobenius representation type* (FFRT) if the indecomposable summands of all the $F_*^e R$ lie in a finite set of isomorphism classes, up to graded shift [15]. Such rings have especially tractable Frobenius behavior. Indeed, once the multiplicity of each summand in the pushforward of each other is known, the F -signature

$$s(R) := \lim_{e \rightarrow \infty} \frac{a_e(R)}{p^{e \dim R}}$$

[4, 16], and the asymptotic proportions of the other summands follow from a Perron eigenvector calculation [12, 18]. In practice, however, determining the multiplicities is considerably harder than identifying the summands themselves, and this information is needed for the F -signature calculation.

The Plücker coordinate ring

$$C_n = k[\mathrm{Gr}(2, n)] = \mathrm{Sym}(k^n \otimes k^2)^{\mathrm{SL}_2}, \quad \dim C_n = 2n - 3,$$

is the fundamental example. Raedschelders, Špenko and Van den Bergh prove that it has FFRT for $p \geq \max\{n - 2, 3\}$ [10, 11]; as they note, this is the first non-trivial example of FFRT for invariants of a group that is not linearly reductive. They identify a family $\mathcal{G}_n$ of $(n - 2) + (n - 3)^2$ indecomposable modules, which consists of the modules of covariants $S\{j\}$, $0 \leq j \leq n - 3$ and certain syzygy modules $K\{j, k\}$, $1 \leq j, k \leq n - 3$, and they prove that $F_*^e C_n$ contains exactly the $S\{j\}$ and the $K\{j, j\}$, while every member of $\mathcal{G}_n$ occurs in $F_*^e C_n$ for $e \geq 2$ [11, Theorems 1.5–1.6]. Their methods determine which summands occur, but outside of the first pushforward of R [10,

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Theorem 15.1], they do not determine how often. Our first main result completes this picture by providing these missing multiplicities.

For each admissible prime p , let $A_n(p)$ be the multiplicity matrix on this family, so the entry $A_n(p)_{M',M}$ counts copies of M' in F_*M with all grading shifts counted together. In particular, rows index the summands being counted, and columns index the modules being pushed forward.

Theorem 1.1. *For any fixed $n \geq 4$, every entry of $A_n(p)$ agrees with a polynomial in $\mathbb{Q}[p]$ of degree at most $2n - 3$ for every prime $p \geq \max\{n - 2, 3\}$.*

The entries are given by nonrecursive finite formulas whose coefficients and summation ranges are independent of p . Section 3 and Appendix A reconstruct the matrix from three counting functions, while Section 4 evaluates those functions by polynomial formulas and verifies the characteristic range and degree bound. This polynomial dependence across characteristics does not follow from FFRT alone. For instance, the Veronese subring $R = k[x, y]^{(3)}$ has FFRT for $p \neq 3$, but the free rank of F_*R is $(p^2 + 2)/3$ when $p \equiv 1 \pmod{3}$ and $(p^2 - 1)/3$ when $p \equiv 2 \pmod{3}$, so no single polynomial in p gives it at every prime.

For the quadric cone C_4 , whose family consists of C_4 and its two spinor modules $S\{1\}$ and $K\{1, 1\}$, we find, up to graded shifts, that

$$F_*C_4 \cong C_4^{\oplus p(13p^4+2)/15} \oplus S\{1\}^{\oplus p(p^4-1)/30} \oplus K\{1, 1\}^{\oplus p(p^4-1)/30},$$

so that for instance $F_*C_4 \cong C_4^{\oplus 211} \oplus S\{1\}^{\oplus 8} \oplus K\{1, 1\}^{\oplus 8}$ when $p = 3$. The full matrix $A_4(p)$ is given in Section 3.5. Since C_4 is the homogeneous coordinate ring of the smooth quadric $\text{Gr}(2, 4) \subset \mathbb{P}^5$, these multiplicities are also determined by Achinger's description of the higher Frobenius push-forwards of line and spinor bundles on quadrics [1], which extends Langer's description of the first push-forward [8]. Their expressions are in terms of dimensions of graded pieces of explicit Artinian modules, and our formulas agree with them.

The formulas underlying Theorem 1.1 determine all Frobenius multiplicities of C_n , with internal shifts counted together, and thus allow us to calculate F -signatures. While a closed form for $s_p(C_n)$ for arbitrary n is not given in our work, we do give the explicit form for the limit F -signature $\lim_{p \rightarrow \infty} s_p(C_n)$. To present this result, we let B_{2m} denote the Bernoulli numbers.

Theorem 1.2. *Suppose $n \geq 4$ and $p \geq \max\{n - 2, 3\}$.*

- (1) *For each $M \in \mathcal{G}_n$ there is a rational function $v_M \in \mathbb{Q}(x)$ such that the multiplicity of M in $F_*^e C_n$ is $v_M(p) p^{e(2n-3)} + o(p^{e(2n-3)})$ as $e \rightarrow \infty$. In particular $s_p(C_n) = v_{C_n}(p)$ is a single rational function of p throughout this range.*
- (2) *The F -signature is determined by a linear system in $\binom{n-2}{2}$ unknowns, obtained from $\lfloor n/2 \rfloor$ Riemann–Roch eigenrows of the multiplicity matrix.*
- (3) *As $p \rightarrow \infty$ through primes, we have*

$$\lim_{p \rightarrow \infty} s_p(C_n) = -\frac{2}{(2n-3)!} \sum_{k=0}^{n-1} (-1)^k \binom{2n}{k} (n-k)^{2n-3} - (2^n - 2) \frac{2^n(2^{2n-2} - 1)(-1)^n B_{2n-2}}{(2n-2)!}.$$

Part (3) is the leading coefficient of the free multiplicity $A_n(p)_{C_n, C_n}$, but this identification is not formal, as a polynomial matrix can have a normalized Perron vector whose limit is not read off from its leading terms. Since C_n is Gorenstein, we obtain this identification from the socle formula for splitting numbers and Smirnov's uniform Hilbert–Kunz estimate; see Section 5. The key point is to show that $\lim_{p \rightarrow \infty} s_p(C_n) = \lim_{p \rightarrow \infty} \frac{a_1(C_n)}{p^{\dim C_n}}$.

For $n = 4, \dots, 8$ the limits are

$$\frac{13}{15}, \quad \frac{97}{126}, \quad \frac{15067}{22680}, \quad \frac{2983}{5280}, \quad \frac{11677453}{24324300}.$$

For C_4 , we calculate $a_e(C_4)$ explicitly, and we provide closed form formulae for $s_p(C_4)$ for every odd prime:

Theorem 1.3. *For every prime $p \geq 3$, we have*

$$s_p(C_4) = \frac{13p^2 + 8}{5(3p^2 + 2)},$$

and

$$a_e(C_4) = \frac{13p^2 + 8}{5(3p^2 + 2)} p^{5e} + \frac{2(p^2 + 1)}{5(3p^2 + 2)} \left(\frac{p(p^2 + 2)}{3} \right)^e.$$

The F -signatures of C_5 and C_6 are rational functions of p of degrees 9/9 and 30/30, given in Section 7. For instance

$$s_p(C_6) = \frac{15067}{22680} - \frac{899}{48600p^4} + O(p^{-5}).$$

The limit 13/15 of $s_p(C_4)$ agrees with Shideler's formula for the limit F -signature of a sum of six squares [13]; see also [2]. Theorem 6.4 of [2], applied to six squares, already gives a free-rank sequence of the form $s_p(C_4)p^{5e} + (1 - s_p(C_4))B^e$, with integers B, C such that $s_p(C_4) = -C/(p^5 - B)$. There B is the coefficient of $\lambda_{(p-1)/2}$ in $\lambda_{(p-1)/2}^5$ in the Han–Monsky representation ring, and the authors note that B and C are hard to determine explicitly when $B > 1$. Theorem 1.3 identifies

$$B = \frac{p(p^2 + 2)}{3}, \quad C = -\frac{p(p^2 - 1)(13p^2 + 8)}{15}.$$

The signature $s_p(C_4)$ is also determined in principle by Achinger's description, through the Perron vector of the resulting 3×3 matrix, but we have not found this closed expression for fixed p in the literature. For $n \geq 5$ the ring C_n is not a hypersurface, and to our knowledge no F -signature of C_n with $n \geq 5$ was previously known.

The method. As noted above, Theorem 1.1 does not follow from FFRT alone. Its proof takes as input the structural results of [11], in particular, the Frobenius-kernel invariants of covariant modules [11, Proposition 5.1, Lemma 5.4], the decomposition theorem [11, Theorem 4.3, Proposition 9.1], the single surviving class of their spectral sequence [11, Lemma 7.8], and induction from Borel invariants [11, Proposition 8.2]. It adds two things: an exact count of each multiplicity, and a proof that these counts are fixed finite sums of polynomials in p . Write Q for the truncated polynomial ring $\text{Sym}(k^n \otimes k^2)/(x_i^p, y_i^p)$. There are three mechanisms. Here a *source* is the module being pushed forward, and a *target* is a summand whose multiplicity is counted.

- (1) For a covariant source $S\{j\}$, the covariant targets are read off from the tilting character of $T(j) \otimes Q$. This character determines the relevant invariants since the module is tilting. The syzygy targets are fusion counts in the level- $(p - 2)$ fusion ring of SL_2 .
- (2) For a syzygy source $K\{j, k\}$, the syzygy targets are dimension counts on the spectral sequence of [11]. After fixing a tuple and a torus weight, its E_3 page is concentrated in one column with at most one surviving class, so no rank of a map is ever needed.
- (3) The covariant targets of a syzygy source require a different calculation. The natural module $K_{jk} \otimes_S Q$ is not known to be tilting, so we cannot determine the count from its character. Instead, [11, Proposition 8.2] expresses the relevant invariants as being induced from Borel invariants. We reconstruct the character of those Borel invariants from Tate cohomology, by Jordan-block arithmetic over the distribution algebra $k[f]/(f^p)$ of the first Frobenius kernel of the unipotent radical.

In each case the multiplicities are finite sums of three counting functions (a weight multiplicity in Q , a fusion count, and a bounded composition count) evaluated at arguments affine in p . The last step is to show that these are polynomials in p throughout the range. For the fusion counts this

uses a Verlinde-type evaluation in terms of sums of powers of cosecants, and for the others the vanishing of binomial coefficients $\binom{x+r}{r}$ at $-r \leq x \leq -1$.

Part (2) of Theorem 1.2 comes from Riemann–Roch on the punctured cone over $\mathrm{Gr}(2, n)$. Its rational Chow ring is $\mathbb{Q}[u]/(u^{\lfloor n/2 \rfloor})$, and Frobenius acts on the component of codimension $2i$ by $p^{2n-3-2i}$, giving $\lfloor n/2 \rfloor$ eigenrows of the multiplicity matrix that are independent of p .

The same geometry leads to the full characteristic polynomial (Theorem 6.5), with the Riemann–Roch factor being complemented by two residual determinants, obtained by splitting formal resolution relations under duality. Both residual matrices are constructed from the bounded-composition counts alone and have entries of degree at most $n - 1$ in p . Thus their construction avoids first assembling the full multiplicity matrix, while retaining the entire characteristic polynomial.

The general theorems are proved by hand, while the rational functions for C_5 and C_6 are computer-assisted, using an accompanying program for some necessary linear algebra calculations, with no interpolation. The program is available as an ancillary file.

In the sequel we apply these results to the generic determinantal rings $k[X_{a \times b}]/I_3(X)$, which are weight-zero parts of the rings C_n for a one-dimensional torus. In the corresponding characteristic range, FFRT descends from the Grassmannian ring by [15, Proposition 3.1.6]. The sequel then determines the minimal summand families and their multiplicities, and computes F -signatures, including that of the 3×3 generic determinantal hypersurface.

We now describe the structure of this work. Section 2 recalls the FFRT formalism. Section 3 derives the multiplicity formulas, with $\mathrm{Gr}(2, 4)$ as a running example, and Section 4 evaluates the counting functions by polynomial formulas, completing the proof of Theorem 1.1. Section 5 proves the characteristic-limit formula in Theorem 1.2(3), while Section 6 gives the Riemann–Roch reduction of Theorem 1.2(2) and the characteristic-polynomial construction of Theorem 6.5. Explicit signatures are given in Section 7. Finally, the boundary specialization of the reconstruction formulas appears in Appendix A.

Preliminaries and conventions. All modules are finitely generated unless otherwise stated. The grading on $F_*^e M$ is divided by p^e . Frobenius summands are considered up to internal shift, and a multiplicity counts all shifts of that type. We retain shifts in the resolutions defining the modules. The number $a_e(R)$ counts all graded free summands of $F_*^e R$, with their internal shifts counted together, while rank denotes rank over the fraction field. Throughout, we let $n \geq 4$, let $p \geq \max\{n - 2, 3\}$, and let

$$C_n = k[\mathrm{Gr}(2, n)] = \mathrm{Sym}(k^n \otimes k^2)^{\mathrm{SL}_2}$$

be the Plücker coordinate ring, which has dimension $d_n = 2n - 3$. The coordinate ring C_n has its standard Plücker grading, while in the invariant presentation below a Plücker generator has polynomial degree two. These gradings differ only by this normalization. Displayed shifts in the equivariant constructions and their invariant modules use the polynomial grading.

For the representation-theoretic calculations, put $V = k^2$, $F = k^n$, $G = \mathrm{SL}(V)$, and $S = \mathrm{Sym}(F \otimes V) = k[x_i, y_i]_{i=1}^n$. Let B be the lower triangular Borel, $\mathbb{T}$ its diagonal torus, and U its unipotent radical. A subscript 1 denotes the kernel of the first Frobenius morphism, as in G_1, B_1, U_1 . We identify torus weights with integer multiples of the fundamental weight ω , with x_i of weight $+\omega$ and y_i of weight $-\omega$. A one-dimensional B -module of weight $j\omega$ is denoted by $(j\omega)$, while Ind_B^G denotes rational induction.

The symbols $L(j)$, $\nabla(j)$ and $T(j)$ denote the simple, induced and indecomposable tilting G -modules of highest weight $j\omega$. We note that for $0 \leq j \leq p - 1$, all three agree with $\mathrm{Sym}^j V$. The character $\mathrm{ch} M$ records torus-weight dimensions, and $\chi_j = \mathrm{ch} \nabla(j) = z^j + z^{j-2} + \cdots + z^{-j}$. The superscript [1] denotes Frobenius twist. We use ordinary duals: $M^\vee = \mathrm{Hom}_A(M, A)$ for a module over a ring A , and $W^* = \mathrm{Hom}_k(W, k)$ for a representation.

Write

$$S_+ = S/(y_1, \dots, y_n), \quad Q = S/S_{>0}^p S = S/(x_1^p, y_1^p, \dots, x_n^p, y_n^p).$$

Here $S_{>0}^p$ is the positive-degree ideal of the subring of p th powers. For $0 \leq j \leq n-3$, the covariant modules are $S\{j\} = (T(j) \otimes_k S)^G$, with $S\{0\} = C_n = S^G$. For $1 \leq j \leq n-3$, put $M_j = \text{Ind}_B^G((j\omega) \otimes_k S_+)$. In our characteristic range, the equivariant graded free resolution [11, Remark 3.4 and (3.5)] is

$$(1) \quad 0 \longrightarrow E_{j,n-1} \longrightarrow \cdots \longrightarrow E_{j,0} \longrightarrow M_j \longrightarrow 0,$$

where

$$E_{j,a} = \begin{cases} T(j-a) \otimes_k \wedge^a F \otimes_k S(-a), & 0 \leq a \leq j, \\ T(a-j-1) \otimes_k \wedge^{a+1} F \otimes_k S(-a-1), & j < a \leq n-1. \end{cases}$$

Using this resolution, define

$$K_{jk} = \Omega^{k+1} M_j = \ker(E_{j,k} \longrightarrow E_{j,k-1}) \quad \text{and} \quad K\{j, k\} = (K_{jk})^G, \quad \text{for } 1 \leq j, k \leq n-3.$$

Thus K_{jk} is an equivariant S -module and $K\{j, k\}$ is a C_n -module. The choice of resolution is part of the definition. We write H^i for rational group-scheme cohomology and $\hat{H}^i$ for Tate cohomology. Further notation from [11] is introduced where it is used.

2. THE FFRT MATRIX AND ITS PERRON EIGENSPACE

We recall the classical FFRT method for encoding Frobenius as a matrix (see [15, Proposition 3.3.1] and [18, Section 3]).

Theorem 2.1. *Let R be a strongly F -regular F -finite positively graded domain of dimension d with perfect degree-zero field and finite F -representation type. Let $M_0 = R, M_1, \dots, M_{h-1}$ represent the distinct indecomposable graded summands occurring in some $F_*^e R$, up to internal shifts. Write*

$$F_* M_j \cong \bigoplus_i M_i^{\oplus A_{ij}} \quad \text{and} \quad \rho = (\text{rank } M_0, \dots, \text{rank } M_{h-1}),$$

with graded shifts suppressed. Then $\rho A = p^d \rho$, the eigenvalue p^d is simple, and its positive right eigenvector v , normalized by $pv = 1$, satisfies

$$\lim_{e \rightarrow \infty} p^{-de} A^e = v\rho \quad \text{with} \quad s(R) = v_0.$$

In particular, with $\chi_A(t) = \det(tI - A)$, we have

$$(2) \quad s(R) = \frac{\text{adj}(p^d I - A)_{00}}{\chi'_A(p^d)}.$$

Proof. The family is Frobenius-closed because each M_j occurs in an iterate of R . Smith–Van den Bergh’s result and its proof show that A is primitive and give the Perron convergence [15, Proposition 3.3.1]. Taking ranks gives $\rho A = p^d \rho$, so this is the Perron eigenvalue and the stated normalization identifies the limit as $v\rho$. Since $(A^e)_{00}$ is the free multiplicity of $F_*^e R$ and $\rho_0 = 1$, its normalized limit is $v_0 = s(R)$. Finally, the adjugate at the simple root p^d is $\text{adj}(p^d I - A) = \chi'_A(p^d) v\rho$, so its $(0, 0)$ -entry gives the formula. $\square$

Corollary 2.2. *Suppose the family and rank vector are fixed and $A(p)$ is the evaluation of a matrix over $\mathbb{Q}[x]$ at every prime in an unbounded range. Then $s(R_p)$ is a single rational function of p on that range, with a finite rational limit as $p \rightarrow \infty$. Every matrix entry has degree at most d .*

Proof. Use (2) and $0 < s(R_p) \leq 1$. Nonzero entries of $A(x)$ have positive leading coefficient, so the rank identity $\rho A(x) = x^d \rho$ bounds their degrees. $\square$

The Plücker rings considered below are Gorenstein. In this case we use $M^\vee = \text{Hom}_R(M, R)$ as noted in Section 1. Finite Frobenius duality identifies $(F_*M)^\vee$ with $F_*(M^\vee)$ up to an internal shift, so in particular, duality permutes the Frobenius summand types, commutes with A , and fixes its normalized Perron vector. Moreover, free summands are invariant under this duality. Let H record total multiplicities on duality orbits. That is, for each source representative, sum the entries of A over a target orbit. Its normalized Perron vector records the total limiting multiplicity on each orbit, and

$$(3) \quad s(R) = \frac{\text{adj}(p^d I - H)_{00}}{\chi'_H(p^d)}.$$

For an arbitrary source, its multiplicity column need not be constant on target duality orbits. The multiplicities in $F_*^e R$ are invariant under duality, since the source R is self-dual, and the limiting vector is invariant as well. Polynomial dependence alone does not identify the characteristic limit with the leading coefficient of A_{00} . That identification follows from the Gorenstein socle formula and uniform Hilbert–Kunz convergence in Section 5.

3. FROBENIUS PUSHFORWARD ON THE PLÜCKER RING

Throughout this section internal grading shifts are counted but suppressed.

As identified in [10, 11], the family of Frobenius summand types over C_n is

$$\mathcal{G}_n = \{S\{j\} : 0 \leq j \leq n-3\} \cup \{K\{j, k\} : 1 \leq j, k \leq n-3\} \quad \text{with} \quad S\{0\} = C_n.$$

By [11, Theorems 1.5–1.6], the types in $F_* C_n$ are the $S\{j\}$ and $K\{j, j\}$, while every member of $\mathcal{G}_n$ occurs in $F_*^e C_n$ for every $e \geq 2$. There are therefore $2n-5$ types at the first iterate and $(n-2) + (n-3)^2$ thereafter. Recall $A_n(p)_{M', M}$ denotes the multiplicity of the target M' in the Frobenius pushforward of the source M .

The ranks of $K\{j, k\}$ equal the S -ranks of K_{jk} . Indeed, on the full-rank matrix locus, taking invariants is descent along an SL_2 -torsor over the punctured cone. Taking alternating ranks in the tail of (1) ending in K_{jk} gives

$$(4) \quad \text{rank } S\{j\} = j+1 \quad \text{and} \quad \text{rank } K\{j, k\} = \begin{cases} (j-k+1)\binom{n-2}{k} + (j-k)\binom{n-2}{k-1}, & k \leq j \\ (k-j+1)\binom{n-2}{k} + (k-j)\binom{n-2}{k+1}, & k \geq j. \end{cases}$$

Note the two expressions for $\text{rank } K\{j, k\}$ agree when $j = k$. The goal of this section will be to compute $A_n(p)$.

3.1. Construction of the matrix. We construct $A_n(p)$ one column at a time, using the ordered family

$$S\{0\}, \dots, S\{n-3\}, \quad K\{1, 1\}, \dots, K\{1, n-3\}, K\{2, 1\}, \dots, K\{n-3, n-3\},$$

with the pairs ordered lexicographically as above. The following recipe determines every entry, while the subsequent subsections justify its steps.

- (1) *Evaluate the counting functions.* The construction uses three functions: $W_n(w)$ counts weights in the truncated polynomial ring Q , $H_{n,q}$ is a fusion multiplicity, and $E_{j,q,e}$ counts bounded compositions of $(q+e)p - j - 2e$ into n parts. Their definitions are given in (12), (17) and (36). To compute symbolically in p , use the binomial formula (13) for W and E , and the finite Bernoulli-number formula (17) for H , with coefficients given by (14) and (15). All ranges depend only on n .
- (2) *Fill a covariant-source column $S\{j\}$.* Form the Weyl coefficients $a_\lambda^{(j)}$ from W_n using (5). Equation (6) gives the free row, and the finite alternating sum (7) gives every other covariant row. Equation (8) places $H_{n,j}$ in every diagonal syzygy row $K\{r, r\}$ and zero in every off-diagonal row. This completes the column.

- (3) *Fill a syzygy-source column $K\{j, k\}$ with $k \leq j$.* First compute the generic term $B_{j,k}$ from the fusion counts using (37). Combine it with the E -counts in (38) and (39) to obtain the Tate coefficients $\mathcal{T}$ and the ordinary-cohomology coefficients $\mathcal{O}$. The syzygy rows are already determined:

$$A_{K\{r,h\},K\{j,k\}} = \mathcal{O}_{j,k;h,r}.$$

To obtain the covariant rows, form P from $\mathcal{T}$ and W_n using (40), and then form Z using the finite sum (41). The difference $Z_{j,k;l} - Z_{j,k;-l-2}$ is the coefficient after Euler induction, before removing the syzygy summands. Subtract their resolution characters, with multiplicities $\mathcal{O}$, exactly as in (43). The result is $A_{S\{l\},K\{j,k\}}$ for each $0 \leq l \leq n-3$.

- (4) *Fill the remaining columns by duality.* If $k > j$, the dual source is $K\{n-2-j, n-2-k\}$, whose second index is smaller than its first. Its column was computed in the previous step. Equation (9) then gives the desired column by fixing the covariant rows and replacing each syzygy row $K\{r, h\}$ by $K\{n-2-r, n-2-h\}$.

For covariant sources, tilting characters determine the covariant part, while fusion counts determine the syzygy part [11, Proposition 5.1]. For syzygy sources, the $\mathcal{O}$ -counts come from a spectral sequence with a single surviving class. The covariant part requires the additional reconstruction through P and Z , as the character of $K_{jk} \otimes_S Q$ alone is not known to determine its decomposition. Instead, Borel invariants and Euler induction recover a character from which the known syzygy contributions can be subtracted (§3.4). Example 3.5 serves as an illustration of the matrix construction, and carries out both kinds of column calculations for $n = 4$.

3.2. Covariant sources. Fix a source $S\{j\}$. We first count weights in $T(j) \otimes Q$, convert them into Weyl-character coefficients, and use finite alternating sums to recover the covariant multiplicities. The two-term tilting characters below explain these sums. A cohomological correction is needed at the free target; fusion counts give the diagonal syzygy targets. Write $C_{r,p}(M) = [x^M](1+x+\dots+x^{p-1})^r$ and $W_n(w) = C_{2n,p}(n(p-1)+w)$ for the multiplicity of the SL_2 -weight w in $Q = S/S_{>0}^p S$. Reflecting the y -exponents converts the weight into a total degree. Let $H_{n,q}(p) = [V_q](\sum_{i=0}^{p-2} V_i)^n$ be the fusion count in the level- $(p-2)$ SL_2 fusion ring, with the labels $p-1$ and -1 interpreted as zero. For a covariant source, [11, Proposition 5.1 and Lemma 5.4] reduce the calculation to the tilting character of $T(j) \otimes Q$, which determines the tilt-free part because $T(j) \otimes Q$ is tilting, and to fusion counts for the higher G_1 -cohomology, which give the K -targets.

For $0 \leq j \leq n-3$, all highest weights in $T(j) \otimes Q$ are at most $n(p-1)+j \leq np-3$. First suppose $p \geq n-1$. In the SL_2 tilting tensor-product formula (see [3]), we have $T(c) = T(c_1) \otimes T(d)^{[1]}$, with $p-1 \leq c_1 \leq 2p-2$. The remaining digit satisfies $0 \leq d \leq n-2 \leq p-1$, and thus $T(d) = \nabla(d)$. The only highest weights relevant to residues 0 and -2 modulo p belong to

$$\Lambda_n = \{0, p-2\} \cup \{ip : 1 \leq i \leq n-1\} \cup \{ip-2 : 2 \leq i \leq n-1\}.$$

Their characters are

$$\mathrm{ch} T(ip) = \chi_{ip} + \chi_{ip-2} \quad \text{and} \quad \mathrm{ch} T(ip-2) = \chi_{ip-2} + \chi_{(i-2)p},$$

apart from the simple weights $0, p-2$.

A Weyl-character coefficient is the difference of consecutive weight multiplicities (whose weights differ by two). For $T(j) \otimes Q$ this difference telescopes to

$$(5) \quad a_\lambda^{(j)} = W_n(\lambda - j) - W_n(\lambda + j + 2), \quad \lambda \in \Lambda_n.$$

The entries of a covariant-source column are the following finite sums:

$$(6) \quad A_{S\{0\}, S\{j\}} = a_0^{(j)} - (2^n - 2)H_{n,j},$$

$$(7) \quad A_{S\{l\}, S\{j\}} = \sum_{r=0}^{\lfloor (n-l-3)/2 \rfloor} (a_{(l+2+2r)p-2}^{(j)} - a_{(l+2+2r)p}^{(j)}) \quad (1 \leq l \leq n-3),$$

$$(8) \quad A_{K\{r,h\}, S\{j\}} = [r=h]H_{n,j}.$$

At a Steinberg source $j = p-1$ all $H_{n,j}$ terms are zero.

To justify these formulas, let $U_\lambda^{(j)}$ be the coefficients obtained by expressing the Weyl character in the two-term characters above, with coefficients beyond the largest index set to zero. For $p \geq n-1$ these are the actual tilting multiplicities. Comparing Weyl coefficients gives

$$U_0^{(j)} + U_{2p-2}^{(j)} = a_0^{(j)} \quad \text{and} \quad U_{ip-2}^{(j)} = a_{ip-2}^{(j)} - a_{ip}^{(j)} + U_{(i+2)p-2}^{(j)} \quad (2 \leq i \leq n-1).$$

Iterating the second identity through the finite range gives exactly the sum in (7), which equals $U_{(l+2)p-2}^{(j)}$. The covariant decomposition of [11, Proposition 5.1 and Lemma 5.4] then gives (6)–(8). Lemma 4.3 verifies the same formulas at $p = n-2$.

3.3. Syzygy targets. Recall that $K_{jk} = \Omega^{k+1}M_j$ is defined using the G -equivariant resolution (1). We also put $L_{jk} = \Omega^{k+1}((j\omega) \otimes S_+)$, using the B -equivariant twisted Koszul resolution of $(j\omega) \otimes S_+$ [11, (3.6)]. Its term in homological degree m is

$$((j-m)\omega) \otimes_k \bigwedge^m F \otimes_k S(-m), \quad 0 \leq m \leq n.$$

Assume first that $k \leq j$. After fixing a tuple and a torus weight, the spectral sequence [11, (6.17), Lemma 7.8] is concentrated in one column at E_3 . Its E_1 dimensions are fusion counts, and the full complex has at most one surviving class, in position $-q$, at tuple degree $(q+e)p - j - 2e$ and row parity $r \equiv q+e \pmod{2}$. Thus a brutal truncation is computed by alternating the dimensions and correcting for the position of this class. The resulting ordinary-cohomology multiplicities count the K -targets by [11, Proposition 7.6, Corollary 8.4, and Theorem 4.3]. Appendix A writes out these alternating sums and indicators (see (37)–(39)) and the resulting entries (42).

For a source with $k > j$ we use ordinary duality. Since C_n is Gorenstein, $F_* \text{Hom}(M, C_n) \cong \text{Hom}(F_* M, C_n)$ up to internal shift. The equivalence between reflexive C_n -modules and reflexive (G, S) -modules [11, (3.1)] commutes with duals: $\text{Hom}_{C_n}(M^G, C_n) \cong \text{Hom}_S(M, S)^G$. For the covariants this gives $S\{j\}^\vee \cong ((T(j) \otimes_k S)^\vee)^G = (T(j)^* \otimes_k S)^G \cong S\{j\}$, since $T(j)^* = (S^j V)^* \cong D^j V = S^j V$ for $j \leq n-3$ [11, Remark 3.4]. For the syzygy modules, [11, Proposition 3.5(1)] gives $K_{jk}^\vee \cong K_{n-j-2, n-k-2}(n) \otimes_k \wedge^n F^*$ as (G, S) -modules, and $\wedge^n F^*$ is a trivial G -representation, so taking invariants gives $K\{j, k\}^\vee \cong K\{n-2-j, n-2-k\}(n)$. It follows that

$$(9) \quad A_{Y,X} = A_{Y^\vee, X^\vee},$$

which reduces to a source whose second K -index is at most its first.

3.4. Covariant targets of syzygy sources. The covariant targets of a syzygy source need the character of an invariant module rather than of $K_{jk} \otimes_S Q$, which is not known to be tilting, and is supplied by the following lemma. Here z records the SL_2 weight, while all other gradings are retained coefficientwise.

Lemma 3.1 (Jordan-block reconstruction). *Let M be a finite-dimensional representation of $\mathbb{T} \ltimes U_1$, where U_1 is the first Frobenius kernel of the unipotent radical of B . Write $H_i(z) = \text{ch } \hat{H}^i(U_1, M)$, including its cohomological torus weight. Then*

$$(10) \quad \text{ch } M^{B_1} = \left[H_0(z) + \frac{(1 - z^2) \text{ch } M + H_1(z) - H_0(z)}{1 - z^{2p}} \right]_{p\mathbb{Z}},$$

where the bracket retains torus weights divisible by p .

Proof. The distribution algebra $\text{Dist}(U_1) = k[f]/(f^p)$ acts on M , where f lowers torus weight by two. Thus $M^{U_1} = \ker f$, and we may decompose M into Jordan blocks with weight-vector bases. Both characters and Tate cohomology are additive on this decomposition, so we first compute the expression inside the bracket for one block.

Let J be a block of length r , with $1 \leq r \leq p$, and let b be the weight of its one-dimensional socle $\ker(f|_J)$. Its weights are $b, b+2, \dots, b+2r-2$, hence

$$\text{ch } J = z^b(1 + z^2 + \dots + z^{2r-2}) \quad \text{and} \quad \text{ch } J^{U_1} = z^b.$$

The two-periodic complex computing Tate cohomology has alternating maps f and f^{p-1} . With the cohomological torus weights included, as in [11, Lemma 3.7], its characters in degrees zero and one are

$$\text{ch } \hat{H}^0(U_1, J) = \text{ch}(\ker f / f^{p-1} J) \quad \text{and} \quad \text{ch } \hat{H}^1(U_1, J) = z^2 \text{ch}(\ker f^{p-1} / f J).$$

The factor z^2 in degree one compensates for the fact that f lowers weight by two.

If $r < p$, then $f^{p-1} J = 0$ and $\ker f^{p-1} = J$. The two Tate characters are therefore z^b and z^{b+2r} . Since

$$(1 - z^2) \text{ch } J = z^b - z^{b+2r},$$

the numerator of the fraction in (10) is zero for this block. Its contribution to the expression inside the bracket is consequently $z^b = \text{ch } J^{U_1}$.

If $r = p$, the block is free over $k[f]/(f^p)$, so both Tate groups vanish. In this case its contribution is

$$\frac{(1 - z^2) \text{ch } J}{1 - z^{2p}} = \frac{z^b(1 - z^{2p})}{1 - z^{2p}} = z^b,$$

again the character of J^{U_1} . Summing over the blocks shows that the expression inside the bracket is $\text{ch } M^{U_1}$. Finally, B_1 is the semidirect product of U_1 with the first Frobenius kernel $\mathbb{T}_1$ of the torus. A torus weight restricts trivially to $\mathbb{T}_1$ precisely when it is divisible by p . Taking these weights gives $\text{ch } M^{B_1}$ and proves the formula. $\square$

Now fix $k \leq j$ and apply the lemma in each finite internal degree of L_{jk} . We retain the internal grading throughout the character calculation and specialize it only after forming finite Hilbert numerators. The expressions below display the resulting weight characters, with z still recording the torus weight.

First, we introduce the two factors

$$\mathcal{K} = \sum_{m=k+1}^n (-1)^{m-k-1} \binom{n}{m} z^{j-m} \quad \text{and} \quad \Sigma = (1 - z^{-p})^n = \frac{\text{ch}(S_+^p)}{\text{ch } S^p}.$$

Then the twisted Koszul resolution defining L_{jk} in §3.3 gives $\text{ch } L_{jk} = \mathcal{K} \text{ch } S$, and since $\text{ch } S / \text{ch } S^p = \text{ch } Q$, the reduced character of L_{jk} is therefore $\mathcal{K} \text{ch } Q$.

For $\ell = 0, 1$, let

$$\mathcal{T}_\ell(z) = \sum_r \mathcal{T}_{j,k;\ell,r} z^{pr}$$

be the finite numerator of the projected Tate character, so that

$$\frac{[\text{ch } \widehat{H}^\ell(U_1, L_{jk})]_{p\mathbb{Z}}}{\text{ch } S^p} = \mathcal{T}_\ell(z)\Sigma.$$

The coefficients $\mathcal{T}_{j,k;\ell,r}$ are the Tate counts calculated by the alternating sums in (38). The polynomial factor before division is $\text{ch}(S_+^p)$. Define Z^{pre} to be the reduced character of the Borel invariants, before induction to $G^{(1)}$. The lemma now gives

$$Z^{\text{pre}} := \frac{\text{ch } L_{jk}^{B_1}}{\text{ch } S^p} = \left[\mathcal{T}_0\Sigma + \frac{(1-z^2)\mathcal{K} \text{ch } Q + (\mathcal{T}_1 - \mathcal{T}_0)\Sigma}{1-z^{2p}} \right]_{p\mathbb{Z}}.$$

All weights of $\text{ch } S^p$ and of the displayed denominator are divisible by p , so they commute with the weight projection. Writing this over a common denominator gives

$$Z^{\text{pre}} = \frac{\mathcal{N}}{1-z^{2p}} \quad \text{and} \quad \mathcal{N} = \left[(1-z^2)\mathcal{K} \text{ch } Q + (\mathcal{T}_1 - z^{2p}\mathcal{T}_0)\Sigma \right]_{p\mathbb{Z}}.$$

As a Hilbert numerator of the finitely generated multigraded S^p -module $L_{jk}^{B_1}$, Z^{pre} is a finite Laurent polynomial, and thus the displayed division is exact. The Tate counts have rows $0 \leq r \leq n-3$ for $\mathcal{T}_0$ and $1 \leq r \leq n-2$ for $\mathcal{T}_1$. Since Σ has normalized weights in $[-n, 0]$, the Tate part of $\mathcal{N}$ has normalized support in $[-(n-1), n-1]$. All weights of $(1-z^2)\mathcal{K} \text{ch } Q$ lie strictly between $-np$ and np , giving the same bounds after projection. Exact division therefore gives

$$(11) \quad \text{supp}_z Z^{\text{pre}}/p \subset [-(n-1), n-3].$$

Write $Z_{j,k;c} = [z^{pc}]Z^{\text{pre}}$ for its normalized weight coefficients. To calculate them, let $P_{j,k;c}$ be the coefficient of z^{pc} in

$$\left[(1-z^2)\mathcal{K} \text{ch } Q + (\mathcal{T}_1 - \mathcal{T}_0)\Sigma \right]_{p\mathbb{Z}}.$$

Expanding $\mathcal{K}$ and Σ gives the finite formula (40). This numerator has no normalized weight below $-n$. In the required range $-(n-1) \leq c \leq n-3$, expansion of $(1-z^{2p})^{-1}$ consequently uses only the terms $\sum_{t=0}^{n-2} z^{2pt}$. Adding the coefficients of $\mathcal{T}_0\Sigma$ gives exactly (41). Thus the character reconstruction requires only the finite sums recorded in Appendix A.

For $k \leq j$, [11, Proposition 8.2, diagram (8.5) and vanishing (8.6)] gives

$$K_{jk}^{G_1} \cong \text{Ind}_{B^{(1)}}^{G^{(1)}} L_{jk}^{B_1} \quad \text{and} \quad R^1 \text{Ind}_{B^{(1)}}^{G^{(1)}} L_{jk}^{B_1} = 0.$$

Hence Euler induction computes the actual character. It sends a normalized weight $c \geq 0$ to χ_c , sends weight -1 to zero, and sends $c \leq -2$ to $-\chi_{-c-2}$. The coefficient of χ_l after induction is therefore $Z_{j,k;l} - Z_{j,k;-l-2}$.

The decomposition in [11, Theorem 4.3 and Proposition 9.1] splits off the syzygy summands already counted by $\mathcal{O}_{j,k;h,r}$, leaving $T \otimes S^p$ for a tilting module T . Subtracting their reduced resolution characters from the induced reduced character gives exactly the exterior-power subtraction in (43). By (11), Euler induction involves only highest weights at most $n-3$, and the reduced syzygy resolutions have the same bound. Thus the residual tilting module has only the simple summands $T(0), \dots, T(n-3)$, since $n-3 \leq p-1$. Its character therefore gives the covariant multiplicities in (43). For $k > j$ we use duality, as in (9).

Ordinary cohomology contributes rows $1 \leq r \leq n-3$. The potential row zero forces tuple degree $-j < 0$, and the potential upper row $n-2$ is excluded by the parity indicator forcing $e = 1$ when $\ell = k$.

3.5. Running example: the quadric cone. For $n = 4$ the family $\mathcal{G}_4$ consists of C_4 , $S\{1\}$ and $K\{1, 1\}$, of ranks 1, 2, 2. Here C_4 is the homogeneous coordinate ring of the smooth quadric $\text{Gr}(2, 4) \subset \mathbb{P}^5$, and $S\{1\}$ and $K\{1, 1\}$ are its two spinor modules. The fusion atoms are

$$H_{4,0} = H_{4,p-2} = \frac{p(p^2 - 1)(p^2 + 1)}{30}, \quad H_{4,1} = H_{4,p-3} = \frac{p(p^2 - 1)(p^2 - 4)}{15},$$

and $2^4 - 2 = 14$. For $n = 4$, the sum in (7) has just one term, $a_{3p-2}^{(j)} - a_{3p}^{(j)}$. For the source C_4 ($j = 0$), the weight counts give

$$a_0^{(0)} = \frac{p(4p^4 - 1)}{3} \quad \text{and} \quad a_{3p-2}^{(0)} - a_{3p}^{(0)} = \frac{p(p^4 - 1)}{30},$$

so (6)–(8) give

$$a_1 = \frac{p(4p^4 - 1)}{3} - 14H_{4,0} = \frac{p(13p^4 + 2)}{15} \quad \text{and} \quad A_{S\{1\}, C_4} = A_{K\{1,1\}, C_4} = \frac{p(p^4 - 1)}{30}.$$

For the source $S\{1\}$, the same calculation gives

$$a_0^{(1)} = \frac{8p(p^2 - 1)^2}{3} \quad \text{and} \quad a_{3p-2}^{(1)} - a_{3p}^{(1)} = \frac{p(p^4 + 10p^2 + 4)}{15}.$$

We now compute the column of $K\{1, 1\}$ directly by step 3 of §3.1.

Write $B = B_{1,1}$ and $E_{q,e} = E_{1;q,e}$. Then Equation (37) gives

$$B = 4H_{4,0} - H_{4,1} = \frac{p(p^2 - 1)(p^2 + 6)}{15}.$$

The only syzygy target is $K\{1, 1\}$, and the alternating terms in (39) cancel to give its multiplicity

$$o := \mathcal{O}_{1,1;1,1} = B + E_{1,0} + E_{3,0} = \frac{p(p^4 + 10p^2 + 4)}{15},$$

since

$$E_{1,0} = \binom{p+2}{3} \quad \text{and} \quad E_{3,0} = \binom{p}{3}.$$

For the second identity, complement the four bounded exponents, so total degree $3p - 1$ becomes $p - 3$.

Next, write w for the normalized weight variable. Equation (38) simplifies to

$$\sum_r \mathcal{T}_{1,1;0,r} w^r = B + E_{3,0} w \quad \text{and} \quad \sum_r \mathcal{T}_{1,1;1,r} w^r = (B + E_{3,0}) w.$$

Substituting these coefficients and the weight counts W_4 into (40)–(41) gives

$$Z_{1,1;0} - Z_{1,1;-2} = 2p^3(p^2 + 1) \quad \text{and} \quad Z_{1,1;1} - Z_{1,1;-3} = -p^3.$$

These are character coefficients before the syzygy subtraction, so negative values are possible. The reduced resolution character of $K\{1, 1\}$ is $4\chi_0 - \chi_1$. Thus (43) gives

$$A_{S\{0\}, K\{1,1\}} = 2p^3(p^2 + 1) - 4o = \frac{2p(p^2 - 1)(13p^2 + 8)}{15}, \quad \text{and}$$

$$A_{S\{1\}, K\{1,1\}} = -p^3 + o = \frac{p(p^2 - 1)(p^2 - 4)}{15}.$$

Together with $A_{K\{1,1\}, K\{1,1\}} = o$, this completes the column. As a check, it is the column of $S\{1\}$ with the two spinor entries exchanged, as required by the outer automorphism $F \leftrightarrow F^*$ of

$D_3 = A_3$, which fixes C_4 and exchanges the spinor modules. Altogether

$$A_4(p) = \frac{p}{15} \begin{pmatrix} 13p^4 + 2 & 2(p^2 - 1)(13p^2 + 8) & 2(p^2 - 1)(13p^2 + 8) \\ \frac{1}{2}(p^4 - 1) & p^4 + 10p^2 + 4 & (p^2 - 1)(p^2 - 4) \\ \frac{1}{2}(p^4 - 1) & (p^2 - 1)(p^2 - 4) & p^4 + 10p^2 + 4 \end{pmatrix},$$

and its rank identity is $(1, 2, 2)A_4(p) = p^5(1, 2, 2)$. Using the matrix $A_4(p)$, we calculate $a_e(C_4)$ for every e , and in turn calculate $s_p(C_4)$ for every odd prime p .

Theorem 3.2. *For every prime $p \geq 3$,*

$$s_p(C_4) = \frac{13p^2 + 8}{5(3p^2 + 2)}, \quad a_e(C_4) = \frac{13p^2 + 8}{5(3p^2 + 2)}p^{5e} + \frac{2(p^2 + 1)}{5(3p^2 + 2)} \left(\frac{p(p^2 + 2)}{3} \right)^e,$$

The signature increases to 13/15 as $p \rightarrow \infty$.

Proof. The matrix commutes with the exchange of the two spinor coordinates. On antisymmetric vectors it acts by p^3 , while on symmetric vectors (v_0, v_1, v_1) it has eigenvalues p^5 and $p(p^2 + 2)/3$. The first row of $A_4v = p^5v$ gives $v_1/v_0 = (p^5 - a_1)/(2A_{C_4, S\{1\}}) = (p^2 + 1)/(2(13p^2 + 8))$, and the rank normalization $v_0 + 4v_1 = 1$ gives $s_p(C_4) = v_0$ by Theorem 2.1. Since $a_e(C_4) = (A_4^e)_{00}$ and the first coordinate vector lies in the invariant subspace of symmetric vectors, $a_e(C_4)$ is a combination of p^{5e} and $(p(p^2 + 2)/3)^e$, determined by $a_0 = 1$ and a_1 . $\square$

4. POLYNOMIAL FORMULAS FOR THE COUNTING FUNCTIONS

Section 3 and Appendix A construct the matrix from the weight counts W_n , the bounded-composition counts $E_{j;q,e}$ and the fusion counts $H_{n,q}$. Here we evaluate these functions by nonrecursive polynomial formulas. The cutoff estimates ensure that the polynomial expression works throughout the stated prime range, rather than only for sufficiently large primes. We then verify the endpoint characteristic and deduce the degree bound, culminating in the proof of Theorem 4.4. For fixed n , these formulas determine $A_n(p)$ over $\mathbb{Q}[p]$ using only finite sums, products, binomial coefficients, and Bernoulli numbers.

4.1. Bounded-composition and weight counts. Recall

$$C_{r,p}(M) = [x^M](1 + x + \cdots + x^{p-1})^r,$$

with a negative coefficient index interpreted as zero. The ordinary SL_2 -weight multiplicity in the truncated polynomial ring $Q = k[x_i, y_i]_{i=1}^n / (x_i^p, y_i^p)$ is

$$(12) \quad W_n(w) = C_{2n,p}(n(p-1) + w).$$

Indeed, reflection of all y -exponents converts the weight $\sum_i (\deg x_i - \deg y_i) = w$ into the displayed total degree.

Lemma 4.1. *Fix $n \geq 4$ and integers α, β . Suppose that either*

$$r = 2n, \quad \text{for } -2n + 1 \leq \beta \leq -1 \quad \text{or} \quad r = n, \quad \text{for } -n + 1 \leq \beta \leq -1.$$

Then, for every integer $p \geq \max\{n - 2, 3\}$, we have

$$(13) \quad C_{r,p}(\alpha p + \beta) = \sum_{\substack{0 \leq t \leq r \\ t < \alpha}} (-1)^t \binom{r}{t} \binom{(\alpha - t)p + \beta + r - 1}{r - 1},$$

where the last binomial coefficient is evaluated as a polynomial in its upper argument. In particular, this is a single polynomial in p , with a summation range independent of p . The two cases cover the weight counts and cohomology counts, respectively, in the construction of $A_n(p)$.

Proof. The identity

$$(1 + x + \cdots + x^{p-1})^r = (1 - x^p)^r (1 - x)^{-r}$$

gives the usual inclusion-exclusion formula

$$C_{r,p}(\alpha p + \beta) = \sum_{\substack{0 \leq t \leq r \\ (\alpha-t)p + \beta \geq 0}} (-1)^t \binom{r}{t} \binom{(\alpha-t)p + \beta + r - 1}{r-1},$$

but its cutoff depends on p . We show that replacing this condition by $t < \alpha$ leaves the value unchanged when the binomial coefficients are interpreted polynomially.

Since $\beta < 0$, every term with $t \geq \alpha$ has $(\alpha-t)p + \beta < 0$ and is excluded from both sums. For $t < \alpha$, put $N = (\alpha-t)p + \beta$. If $N \geq 0$, the term occurs in both sums with the same value. If $N < 0$, it occurs only in (13), and we must show that its polynomial value is zero.

Because $\alpha - t \geq 1$ and $p \geq n - 2$, the hypotheses give

$$N \geq p + \beta \geq \begin{cases} -n - 1 \geq -(2n - 1) = -(r - 1), & r = 2n, \\ -1 \geq -(n - 1) = -(r - 1), & r = n. \end{cases}$$

Thus any such negative N lies in $\{-(r-1), \dots, -1\}$. For these values the polynomial

$$\binom{N + r - 1}{r - 1} = \frac{(N + 1)(N + 2) \cdots (N + r - 1)}{(r - 1)!}$$

vanishes. Every extra term introduced by the fixed cutoff therefore contributes zero, proving (13). Its summands are polynomials in p and its range is fixed, proving polynomiality. $\square$

4.2. Fusion counts. Recall that, with $V_0, \dots, V_{p-2}$ the usual basis of the level- $(p-2)$ SL_2 fusion ring,

$$H_{n,q}(p) = [V_q] \left(\sum_{i=0}^{p-2} V_i \right)^n.$$

The endpoint label $q = p - 1$ and its reflection -1 mean zero. Write $\chi_q(z) = z^q + z^{q-2} + \cdots + z^{-q}$, also regarded as a polynomial in $z + z^{-1}$. For $0 \leq q \leq n - 3$, define integers $b_{n,q,m}$ by

$$(14) \quad (x^{1-n} - x^{2-n}) \chi_q(2 - 4x) = \sum_{m=1}^{n-1} b_{n,q,m} x^{-m}.$$

Here $\chi_q(2 - 4x)$ means the polynomial in $z + z^{-1}$, evaluated at $2 - 4x$. It can be expanded directly using

$$\chi_q(X) = \sum_{h=0}^{\lfloor q/2 \rfloor} (-1)^h \binom{q-h}{h} X^{q-2h}.$$

Recall B_{2j} denotes the Bernoulli numbers, with $B_2 = 1/6$, and define rational coefficients $c_{m,j}$ by the finite product

$$(15) \quad \frac{1}{(2m-1)!} \prod_{h=1}^{m-1} (u + 4h^2) = \sum_{j=0}^{m-1} c_{m,j} u^j \quad (m \geq 1).$$

The empty product is 1, so $c_{1,0} = 1$.

Lemma 4.2. Put $P_m(z) = \sum_{r=1}^{z-1} \csc^{2m}(r\pi/z)$ for positive integral z . Its polynomial extension is

$$(16) \quad P_m(z) = \sum_{j=0}^{m-1} c_{m,j} \frac{(-1)^j 2^{2j+2} B_{2j+2}}{2j+2} (z^{2j+2} - 1).$$

The fusion counts are given by the finite polynomial formula

$$(17) \quad \begin{aligned} H_{n,q}(p) &= \frac{2^{2-n}}{p} \sum_{m=1}^{n-1} b_{n,q,m} (P_m(2p) - P_m(p)) \\ &= 2^{2-n} \sum_{m=1}^{n-1} \sum_{j=0}^{m-1} b_{n,q,m} c_{m,j} \frac{(-1)^j 2^{2j+2} (2^{2j+2} - 1) B_{2j+2}}{2j+2} p^{2j+1}. \end{aligned}$$

In particular $H_{n,q} \in \mathbb{Q}[p]$ has degree at most $2n-3$. Moreover $H_{n,p-2-q} = H_{n,q}$, including the zero endpoint convention.

Proof. At $X_r = 2 \cos(r\pi/p)$, the sum of the fusion characters is

$$\sum_{i=0}^{p-2} \chi_i(X_r) = \frac{1 - (-1)^r}{2 - 2 \cos(r\pi/p)}.$$

The sine-orthogonality projector for V_q , with $x_r = \sin^2(r\pi/(2p))$, therefore gives

$$H_{n,q}(p) = \frac{2^{3-n}}{p} \sum_{\substack{1 \leq r < p \\ r \text{ odd}}} (x_r^{1-n} - x_r^{2-n}) \chi_q(2 - 4x_r).$$

The Laurent polynomial in (14) vanishes at $x = 1$. Reflection $r \mapsto 2p - r$ doubles this sum, and the possible middle term $r = p$ is zero. Separating odd and even indices proves (17). The reflected fusion label has the same sine projector on odd indices, proving the last assertion.

To prove (16), put $\partial = d/dx$ and

$$L_m(\partial) = \prod_{h=1}^{m-1} \frac{\partial^2 + 4h^2}{2h(2h+1)} = \sum_{j=0}^{m-1} c_{m,j} \partial^{2j}.$$

The identity $(\csc^{2m} x)'' + 4m^2 \csc^{2m} x = 2m(2m+1) \csc^{2m+2} x$ shows that $\csc^{2m} x = L_m(\partial) \csc^2 x$. Applying this operator to $\sum_{r=0}^{z-1} \csc^2(x+r\pi/z) = z^2 \csc^2(zx)$ and removing the term $r = 0$ expresses $P_m(z)$ as the constant Laurent coefficient of $L_m(\partial)(z^2 \csc^2(zx) - \csc^2 x)$. The Bernoulli expansion of $\cot x$, differentiated once, gives

$$(2j)! [x^{2j}] \csc^2 x = \frac{(-1)^j 2^{2j+2} B_{2j+2}}{2j+2}.$$

Taking the constant coefficient proves (16). Substitution gives the second line of (17), whose largest possible exponent is $2n-3$. $\square$

4.3. The characteristic range and degree bound. The character calculation in §3.2 first assumed $p \geq n-1$. The following lemma verifies the endpoint $p = n-2$ of the permissible range before we assemble the polynomial formulas.

Lemma 4.3. *The finite formulas (6)–(8) give the needed multiplicities at $p = n-2$, whenever this is an odd prime, including $n = 5, p = 3$.*

Proof. Here $p \geq 3$ and every source $T(j)$, for $0 \leq j \leq n-3 = p-1$, is simple. The bound $n(p-1)+j \leq np-3$ still restricts the relevant linked weights to Λ_n . Write τ_λ for the formal two-term characters used in the tilting inversion. The tilting tensor-product formula gives $\tau_\lambda = \text{ch } T(\lambda)$ on this list with exactly one exception. Put $H = p(p+1)$ and $L = p(p-1)$. Since $T(H) = T(p) \otimes T(p)^{[1]}$, multiplication of $\text{ch } T(p) = \chi_p + \chi_{p-2}$ with its twist gives

$$\text{ch } T(H) = \chi_H + \chi_{H-2} + \chi_L + \chi_{L-2} = \tau_H + \tau_L.$$

Every other relevant weight has remaining tilting digit at most $p - 1$. Consequently the actual multiplicities agree with the formal inverse U_λ except that the coefficient at L is $U_L - U_H$. The covariant entries read only

$$U_0, \quad U_{2p-2}, \quad \text{and} \quad U_{(l+2)p-2} \quad (1 \leq l \leq p-1),$$

so the altered coefficient is unused. The formulas (6)–(7) are exactly these same formal coefficients written as finite sums, so they remain valid. Equivalently, $T(L)^{G_1} = 0$, since its first tilting digit is p and $T(p)^{G_1} = 0$.

For $j \leq p - 2$, the covariant decomposition and fusion correction therefore apply unchanged. At $j = p - 1$, the Steinberg module is G_1 -projective, so higher cohomology vanishes and the decomposition is tilt-free with coefficient module $(T(p - 1) \otimes Q)^{G_1}$ [11, Proposition 5.1 and its proof]. The formulas give the same result: all fusion labels are $p - 1$ or -1 , and $H_{n,p-1} = 0$ by Lemma 4.2.

Finally, for the K-columns, the published cohomology and induction arguments hold for $p \geq n - 2$, while the residual tiltings have highest weight at most $n - 3 = p - 1$ and remain simple. The finite-truncation argument of §3.4 and the cutoff estimates above also hold at this endpoint. $\square$

Finally, we are ready to prove the main result of this section:

Theorem 4.4 (Polynomiality of the multiplicities). *For fixed $n \geq 4$, every entry of $A_n(p)$ agrees with a polynomial in $\mathbb{Q}[p]$ of degree at most $d_n = 2n - 3$ throughout the prime range $p \geq \max\{n - 2, 3\}$. The entries are given by (6)–(8) for covariant sources and by (42)–(43) and (9) for syzygy sources.*

Proof. Section 3 and Appendix A express every entry as a finite linear combination, with coefficients and ranges independent of p , of W_n , $E_{j;q,e}$ and $H_{n,q}$. Every weight query has constant weight in $[-(n - 1), n - 1]$, so its bounded-composition index has constant part in $[-2n + 1, -1]$, while every E -index has constant part $-j - 2e \in [-n + 1, -1]$. Lemma 4.1 therefore supplies a polynomial for each such count throughout the prime range. All fusion labels reduce to $0 \leq q \leq n - 3$, and (17) supplies their polynomials. The support argument in §3.4 makes every coefficient extraction a finite sum with range depending only on n .

Substituting these formulas proves polynomiality of every entry. Section 3 and Lemma 4.3 identify these polynomials with the actual multiplicities at every admissible prime. Finally, the rank identity $\rho A_n(p) = p^{d_n} \rho$ and positivity of the ranks give the degree bound by Corollary 2.2. $\square$

5. CHARACTERISTIC LIMITS

For each fixed prime, Section 2 obtains the signature from the normalized Perron eigenvector. In this section we show $\lim_{p \rightarrow \infty} s_p(C_n) = \lim_{p \rightarrow \infty} \frac{a_1(C_n)}{p^{d_n}}$ and use this to give an explicit calculation for the limit F -signature. Since the Plücker rings are Gorenstein, this comparison follows from the socle formula for splitting numbers and Smirnov's uniform Hilbert–Kunz estimate [14]. The polynomiality established in Section 4 then identifies the limit with a leading coefficient, which we evaluate explicitly.

5.1. Comparison with the first free multiplicity. Fix $n \geq 4$ and recall $d_n := \dim(C_n) = 2n - 3$. For an ideal I primary to the homogeneous maximal ideal of C_n , write

$$e_{\text{HK}}(I) = \lim_{e \rightarrow \infty} \frac{\ell(C_n/I^{[p^e]})}{p^{ed_n}}$$

for its Hilbert–Kunz multiplicity.

Theorem 5.1. *For each fixed $n \geq 4$, we have*

$$(18) \quad s_p(C_n) = \frac{a_1(C_n)}{p^{2n-3}} + O(p^{-1}) \quad \text{and} \quad \lim_{p \rightarrow \infty} s_p(C_n) = [p^{d_n}]a_1(C_n).$$

The implied constant may depend on n , but is independent of p .

Proof. Let $\mathcal{C}_n$ be the integral Plücker algebra. Choose a homogeneous system of parameters $x_1, \dots, x_{d_n}$ in $\mathcal{C}_n \otimes \mathbb{Q}$, and choose a homogeneous element u whose class generates the socle modulo $Q = (x_1, \dots, x_{d_n})$. After inverting a nonzero integer L , these elements are defined over $B = \mathbb{Z}[1/L]$. Enlarging L makes $\mathcal{C}_n/Q$ finite free over B and ensures that the class of u remains a nonzero socle element in every fiber. Indeed, the finitely many equations expressing annihilation by the degree-one generators hold after clearing denominators, and the nonzero class of u can be made part of a B -basis. Each fiber is Gorenstein, so this class generates its socle. Set $J = (Q, u)$.

For $p \nmid L$, put $R_p = \mathcal{C}_n \otimes_B \mathbb{F}_p$, and write Q_p, J_p, u_p for the reductions of these choices. The Gorenstein socle formula [9, Corollary 6.8(i)], which gives the Hilbert–Kunz description of F -signature from [4], yields

$$(19) \quad a_1(R_p) = \ell(R_p/(Q_p^{[p]} : u_p^p)) = \ell(R_p/Q_p^{[p]}) - \ell(R_p/J_p^{[p]}),$$

$$(20) \quad s(R_p) = e_{\text{HK}}(Q_p) - e_{\text{HK}}(J_p).$$

The second equality in (19) follows from multiplication by u_p^p modulo $Q_p^{[p]}$. These formulas also show that the quantities are unchanged upon extending the perfect ground field, so they compute the multiplicity and signature of C_n .

The algebra $\mathcal{C}_n$ is flat over B , with geometrically integral fibers of dimension d_n , and $\mathcal{C}_n/J$ is finite over B . Thus $B \rightarrow \mathcal{C}_n$ with the ideal J is an affine J -family in the sense of [14, Definition 3.8]. Smirnov’s uniform estimate [14, Corollary 4.9] therefore gives, after excluding finitely many further primes,

$$\left| e_{\text{HK}}(J_p) - \frac{\ell(R_p/J_p^{[p]})}{p^{d_n}} \right| \leq \frac{C}{p},$$

where C is independent of p . Since Q_p is generated by a regular sequence,

$$\frac{\ell(R_p/Q_p^{[p]})}{p^{d_n}} = \ell(R_p/Q_p) = e_{\text{HK}}(Q_p).$$

Subtracting (19), divided by p^{d_n} , from (20) proves the comparison. Finally, $a_1(C_n)$ is a polynomial in p of degree at most d_n by Section 4, so its normalized limit is its coefficient of p^{d_n} . $\square$

The comparison itself uses Gorensteinness and uniform Hilbert–Kunz convergence, and does not require FFRT. The Frobenius decomposition supplies the explicit polynomial $a_1(C_n)$ that we will use to evaluate the limit.

5.2. The first free multiplicity and the limit.

Proposition 5.2. *The first free multiplicity is the explicit polynomial*

$$(21) \quad a_1(C_n) = C_{2n,p}(n(p-1)) - C_{2n,p}(n(p-1)+2) - (2^n - 2)H_{n,0}(p).$$

Proof. Set $j = 0$ in (6) and use (5) and (12). The endpoint characteristic is included by Lemma 4.3. $\square$

Let B_{2r} denote the Bernoulli numbers, with $B_2 = 1/6$. Define

$$\gamma_n = \frac{2^n(2^{2n-2} - 1)(-1)^n B_{2n-2}}{(2n-2)!}.$$

Corollary 5.3. *For every fixed $n \geq 4$,*

$$(22) \quad \lim_{p \rightarrow \infty} s(C_n) = -\frac{2}{(2n-3)!} \sum_{k=0}^{n-1} (-1)^k \binom{2n}{k} (n-k)^{2n-3} - (2^n - 2)\gamma_n.$$

In particular, for $n = 4, 5, 6, 7, 8$ the values are respectively

$$\frac{13}{15}, \quad \frac{97}{126}, \quad \frac{15067}{22680}, \quad \frac{2983}{5280}, \quad \frac{11677453}{24324300}.$$

Proof. By Theorem 5.1, the required limit is $[p^{2n-3}]a_1(C_n)$. We compute this coefficient separately for the two contributions in (21).

First consider the bounded-composition terms. Put $a = n(p-1)$. The symmetry $C_{2n,p}(a-2) = C_{2n,p}(a+2)$ gives

$$C_{2n,p}(a) - C_{2n,p}(a+2) = -\frac{1}{2}(C_{2n,p}(a+2) - 2C_{2n,p}(a) + C_{2n,p}(a-2)).$$

For sufficiently large p , formula (13) gives, for $h = -2, 0, 2$,

$$C_{2n,p}(a+h) = \sum_{k=0}^{n-1} (-1)^k \binom{2n}{k} \binom{(n-k)p + n - 1 + h}{2n-1}.$$

To calculate the indicated combination, regard $f(X) = \binom{X}{2n-1}$ as a polynomial in X . Expanding its three translates shows that

$$f(X+2) - 2f(X) + f(X-2) = \frac{4}{(2n-3)!} X^{2n-3} + \text{terms of degree at most } 2n-4.$$

Indeed, the terms in degrees $2n-1$ and $2n-2$ cancel, and the displayed coefficient comes from the leading term $X^{2n-1}/(2n-1)!$ of f . Substituting $X = (n-k)p + n - 1$, multiplying by $-1/2$, and summing gives

$$[p^{2n-3}](C_{2n,p}(a) - C_{2n,p}(a+2)) = -\frac{2}{(2n-3)!} \sum_{k=0}^{n-1} (-1)^k \binom{2n}{k} (n-k)^{2n-3}.$$

Next consider $H_{n,0}(p)$. Since $\chi_0 = 1$, equation (14) has only the two nonzero coefficients $b_{n,0,n-1} = 1$ and $b_{n,0,n-2} = -1$. Thus (17) becomes

$$H_{n,0}(p) = \frac{2^{2-n}}{p} (P_{n-1}(2p) - P_{n-1}(p) - P_{n-2}(2p) + P_{n-2}(p)).$$

The terms involving P_{n-2} have degree at most $2n-5$ after division by p , so they do not contribute to $[p^{2n-3}]H_{n,0}$. By (16) and $c_{n-1,n-2} = 1/(2n-3)!$, the leading coefficient of $P_{n-1}(z)$ is

$$[z^{2n-2}]P_{n-1}(z) = \frac{(-1)^n 2^{2n-2} B_{2n-2}}{(2n-2)!}.$$

Consequently

$$[p^{2n-3}]H_{n,0}(p) = 2^{2-n}(2^{2n-2} - 1) \frac{(-1)^n 2^{2n-2} B_{2n-2}}{(2n-2)!} = \gamma_n.$$

Combining the two coefficients according to (21) proves (22). $\square$

Corollary 5.4. *Write $s_\infty(C_n) = \lim_{p \rightarrow \infty} s(C_n)$. Then*

$$s_\infty(C_n) = 12\sqrt{\frac{3}{\pi}} n^{-3/2} \left(1 - \frac{3}{8n} + O(n^{-2})\right) \quad (n \rightarrow \infty).$$

In particular, $\lim_{n \rightarrow \infty} s_\infty(C_n) = 0$. For illustration, evaluation of (22) gives the following values, rounded to eight decimal places:

n	10	20	50	100
$s_\infty(C_n)$	0.35473508	0.12866237	0.03291929	0.01168254

Proof. Let M_n be the first term on the right of (22), and let f_n be the density of the sum of $2n$ independent uniform random variables on $[-1/2, 1/2]$. The convolution formula for this density gives $M_n = -2f_n''(0)$. Fourier inversion, followed by the change of variable $t \mapsto 2t$, yields

$$M_n = \frac{16}{\pi} \int_0^\infty t^2 \left(\frac{\sin t}{t} \right)^{2n} dt.$$

For every fixed $\delta > 0$, the integral over $[\delta, \infty)$ is exponentially small. Indeed, $|\sin t/t|$ is bounded there by a constant less than one, and $t^2 |\sin t/t|^8$ is integrable. Near zero, $\log(\sin t/t) = -t^2/6 + O(t^4)$. Substituting $t = u/\sqrt{n}$ and using dominated convergence therefore gives

$$\lim_{n \rightarrow \infty} n^{3/2} M_n = \frac{16}{\pi} \int_0^\infty u^2 e^{-u^2/3} du = 12\sqrt{\frac{3}{\pi}}.$$

Keeping one more term, $\log(\sin t/t) = -t^2/6 - t^4/180 + O(t^6)$, the same substitution gives $(\sin t/t)^{2n} = e^{-u^2/3} (1 - u^4/(90n)) + O(n^{-2})$ in the relevant range, with a dominated error. Since

$$\int_0^\infty u^6 e^{-u^2/3} du = \frac{135}{4} \int_0^\infty u^2 e^{-u^2/3} du \quad \text{and} \quad \frac{135}{4} \cdot \frac{1}{90} = \frac{3}{8},$$

this yields the factor $1 - \frac{3}{8n} + O(n^{-2})$. Finally, the Bernoulli–zeta identity $(-1)^n B_{2n-2} = 2(2n-2)! \zeta(2n-2)/(2\pi)^{2n-2}$ shows that

$$(2^n - 2)\gamma_n = O\left(\left(\frac{4}{\pi^2}\right)^n\right),$$

and this correction does not affect the asymptotic. $\square$

Remark 5.5. Over a field of characteristic different from two, C_4 is the cone over a quadric in six variables, which is defined by a sum of six squares. For a sum of $2m$ squares, Theorem 24 and Corollary 22 of [13] give the limit F -signature $1 + (-1)^m 2^{2m} (2^{2m} - 1) B_{2m}/(2m)!$; at $m = 3$ this is $13/15$, the value of (22) at $n = 4$, and we note that the two expressions come from unrelated arguments.

6. RIEMANN–ROCH REDUCTION AND THE FROBENIUS SPECTRUM

We first record the matrix consequence of the standard Frobenius compatibility in Riemann–Roch [5, (2.8)]; see also [6, Section 4] for its use with FFRT summand classes. Here $G_0(Y)$ is the Grothendieck group of coherent sheaves, and $\text{CH}_i(Y)$ is the Chow group of dimension- i cycles.

Proposition 6.1 (Riemann–Roch eigenrows). *Let Y be a d -dimensional quasi-projective variety over a perfect field of characteristic $p > 0$, and let $\mathcal{E}_0, \dots, \mathcal{E}_{h-1}$ be coherent sheaves with $F_* \mathcal{E}_j \cong \bigoplus_i \mathcal{E}_i^{\oplus A_{ij}}$. Let $\tau_i : G_0(Y)_\mathbb{Q} \rightarrow \text{CH}_i(Y)_\mathbb{Q}$ be the dimension- i Riemann–Roch component, and put*

$$V_i = \text{span}_\mathbb{Q} \{ \tau_i([\mathcal{E}_j]) : 0 \leq j < h \}, \quad r_i = \dim_\mathbb{Q} V_i.$$

For a basis $\ell_{i,1}, \dots, \ell_{i,r_i}$ of V_i^ , the rows*

$$w_{i,a} = (\ell_{i,a}(\tau_i([\mathcal{E}_j])))_{j=0}^{h-1} \quad \text{satisfy} \quad w_{i,a} A = p^i w_{i,a}.$$

All $\sum_i r_i$ rows are linearly independent. Thus every i with $r_i > 0$ gives an eigenvalue p^i of geometric multiplicity at least r_i , and different dimensions i give distinct eigenvalues.

Proof. Frobenius compatibility gives $\tau_i(F_*\alpha) = p^i \tau_i(\alpha)$. Applying $\ell_{i,a} \tau_i$ to the decompositions proves the eigenrow identity. For fixed i , evaluation on the spanning set $\{\tau_i([\mathcal{E}_j])\}_j$ embeds V_i^* into $\mathbb{Q}^h$, so these r_i rows are independent. Since the scalars p^i are distinct, the rows from different dimensions are independent as well. $\square$

For smooth Y , the class $\tau([\mathcal{E}])$ is $\text{ch}(\mathcal{E}) \text{td}(T_Y) \cap [Y]$. We now compute these classes on the punctured Plücker cone and prove independence of the resulting eigenrows. This proves Theorem 1.2(2), independently of Section 5. Put

$$d = 2n - 3, \quad q = \lfloor n/2 \rfloor, \quad \text{and} \quad m = n - 2 + \left\lceil \frac{(n-3)^2}{2} \right\rceil.$$

Let $B = B_n(p)$ be the $m \times m$ matrix obtained from $A_n(p)$ by recording total multiplicities on the duality orbits

$$S\{j\}^\vee = S\{j\}, \quad K\{j, k\}^\vee = K\{n-2-j, n-2-k\}.$$

Thus each entry of B is the sum over a target orbit for one source representative, as in (3). Here ρ denotes the rank row on these orbits. The theorem below supplies $q < m$ independent eigenrows, determining part of the spectrum of B .

Theorem 6.2. *Under the hypotheses of Theorem 4.4, B has q linearly independent rational left eigenrows $w_0, \dots, w_{q-1}$, independent of p , satisfying*

$$(23) \quad w_i B = p^{d-2i} w_i, \quad w_0 = \rho.$$

They are given by

$$(24) \quad T_n(x) = \left(\frac{x}{2 \sinh(x/2)} \right)^{2n} \left(\frac{\sinh x}{x} \right)^2,$$

$$(25) \quad w_i(S\{j\}) = [x^{2i}] T_n(x) \sum_{a=0}^j e^{(j-2a)x},$$

$$(26) \quad w_i(K\{j, k\}) = \sum_{a=k+1}^{n-1} (-1)^{a-k-1} \binom{n}{b_a} w_i(S\{c_a\}),$$

where $(c_a, b_a) = (j-a, a)$ for $a \leq j$, and $(c_a, b_a) = (a-j-1, a+1)$ for $a > j$. These values are constant on each duality orbit. Consequently the normalized Perron eigenvector satisfies

$$(27) \quad w_0 v = 1, \quad w_i v = 0 \quad (1 \leq i < q),$$

and $s_p(C_n)$ is determined by a linear system with $m - q = \binom{n-2}{2}$ unknowns.

Proof. Write $X = \text{Gr}(2, n)$ and let $\pi : U \rightarrow X$ be the punctured Plücker cone, the complement of the zero section in $\mathcal{O}_X(-1)$. With $h = c_1(\mathcal{O}_X(1))$, localization for this line bundle gives

$$(28) \quad \text{CH}^*(U)_{\mathbb{Q}} = \text{CH}^*(X)_{\mathbb{Q}}/(h) = \mathbb{Q}[u]/(u^q), \quad \deg u = 2.$$

For the last equality use the rank-two quotient bundle $\mathcal{Q}$: the usual presentation has generators $c_1(\mathcal{Q}), c_2(\mathcal{Q})$ and relations the complete symmetric polynomials of degrees $n-1$ and n . Setting $c_1(\mathcal{Q}) = h = 0$ makes its formal roots $x, -x$, while the relations become $u^q = 0$ with $u = x^2 = -c_2(\mathcal{Q})$. This is the same hyperplane quotient that computes the local cone's Chow groups in [7, Theorem 2.3].

Every module in $\mathcal{G}_n$ restricts to a vector bundle on U , since it occurs in a Frobenius pushforward and U is smooth. For such a module M , define $w_i(M)$ to be the coefficient of u^i in

$$\text{ch}(M|_U) \text{td}(T_U) \in \text{CH}^*(U)_{\mathbb{Q}}.$$

The relative tangent bundle of π is trivial, so $\mathrm{td}(T_U) = \pi^* \mathrm{td}(T_X)$. From the tautological sequence and $T_X = \mathcal{R}^\vee \otimes \mathcal{Q}$ one obtains in $K_0(X)$

$$[T_X] = n[\mathcal{Q}] - [\mathcal{Q}^\vee \otimes \mathcal{Q}].$$

Modulo h , the roots of the second term are $0, 0, 2x, -2x$. The Todd formula therefore gives (24). The bundle associated with $S\{j\}$ is $\mathrm{Sym}^j \mathcal{Q}$, up to grading normalization [11, Proposition 11.2], and its character gives (25).

For completeness, the syzygy formula requires exactness only on U . The full-rank locus of $n \times 2$ matrices is a principal SL_2 -bundle over U . Restrict the equivariant resolution [11, (3.5), (3.7)] to this locus and descend its exact tail resolving $\Omega^{k+1} M_j$. Descent along a torsor is exact, and its term in homological degree a is, after forgetting a graded shift, $\binom{n}{b_a}$ copies of the bundle for $S\{c_a\}$. Graded shifts are trivial after restriction to U . Equivalently, line bundle twists disappear in (28). Taking the alternating class of this tail proves (26). All the symmetric or divided powers involved have degree at most $n - 3 < p$, so this uses precisely the stated characteristic bound. No exactness of the global invariant functor is being asserted.

Applying Proposition 6.1 on U to the dimension- $(d-2i)$ components gives (23) before aggregation. Ordinary module duality restricts to bundle duality on U . Only even codimensions survive in (28), and bundle duality fixes these Chern-character components. Hence the rows are constant on duality orbits and descend to B .

Finally,

$$\sum_{a=0}^j e^{(j-2a)x} = \frac{\sinh((j+1)x)}{\sinh x}$$

shows that $w_i(S\{j\}) = (j+1)P_i((j+1)^2)$, where P_i has degree i and leading coefficient $1/(2i+1)!$. The minor on the columns $S\{0\}, \dots, S\{q-1\}$ is consequently

$$\det(w_i(S\{j\}))_{0 \leq i, j < q} = \frac{\prod_{a=1}^q a \prod_{\substack{1 \leq a < b \leq q \\ q-1}} (b^2 - a^2)}{\prod_{i=0}^{q-1} (2i+1)!} \neq 0,$$

which proves independence. Applying the rows to $Bv = p^d v$ and using rank normalization gives (27). Eliminating these q covariant coordinates gives the asserted number of unknowns. $\square$

Here is an explicit form of the resulting smaller system. Let W have rows w_i , let E index $S\{0\}, \dots, S\{q-1\}$, and let J index the remaining duality orbits. Partition $W = (W_E \ W_J)$ and B accordingly, and put $e = (1, 0, \dots, 0)^t \in \mathbb{Q}^q$. Set

$$(29) \quad H = B_{JJ} - B_{JE} W_E^{-1} W_J, \quad b = B_{JE} W_E^{-1} e, \quad M = p^d I - H.$$

Then the coordinates $z = v_J$ satisfy

$$(30) \quad Mz = b, \quad s_p(C_n) = e^t W_E^{-1} (e - W_J z).$$

Indeed $v_E = W_E^{-1} (e - W_J z)$, and the J -rows of the eigenvector equation give (29). The other rows follow from $WB = \mathrm{diag}(p^{d-2i})W$ and the invertibility of W_E . In coordinates (Wv, v_J) the matrix B is block triangular, and so

$$(31) \quad \det(tI - B) = \prod_{i=0}^{q-1} (t - p^{d-2i}) \det(tI - H).$$

The simplicity of the Perron root implies $\det M \neq 0$. Thus (30), or Cramer's rule applied to it, is an exact uniform formula of size $\binom{n-2}{2}$. This gives sizes 1, 3, 6 for $n = 4, 5, 6$. It does not assert

that this size is minimal or supply a scalar evaluation of the remaining determinant for arbitrary values of n .

6.1. The full characteristic polynomial. The quotient calculation above accounts for q eigenvalues. We now construct the remaining characteristic factors directly from the bounded-composition atoms $E_{j;s,e}$ of (36), without first assembling $A_n(p)$. Put $\nu = n - 3$ and let $\mathcal{V}_n$ be the rational vector space freely generated by $e_a = e_{S\{a\}}$ and $f_{jk} = e_{K\{j,k\}}$. Let D be the linear involution induced by module duality: it fixes the e_a and sends f_{jk} to $f_{\bar{j},\bar{k}}$, where $\bar{j} = \nu + 1 - j$. By (9), it commutes with $A_n(p)$.

Using the resolution (1) and its indices (c_a, b_a) in (26), set

$$(32) \quad t_{jk} = \sum_{a=k+1}^{n-1} (-1)^{a-k-1} \binom{n}{b_a} e_{c_a}, \quad x_{jk} = f_{jk} - t_{jk} \quad (1 \leq j, k \leq \nu).$$

Exactness of the tail on the punctured cone, proved in Theorem 6.2, places every x_{jk} in $\mathcal{N}_n = \ker(\mathcal{V}_n \rightarrow K_0(U)_{\mathbb{Q}})$. These are formal signed classes, and their coordinates need not be nonnegative.

Lemma 6.3. *The duality eigenspaces $\mathcal{N}_n^+$ and $\mathcal{N}_n^-$ have bases*

$$\begin{aligned} \mathcal{B}^+ &= (y_{jk} : 1 \leq k \leq j \leq \nu), & y_{jk} &= \frac{1}{2}(x_{jk} + Dx_{jk}), \\ \mathcal{B}^- &= (a_{jk} : 1 \leq k < j \leq \nu) \cup (a_{jj} : 1 \leq j \leq \lfloor \nu/2 \rfloor), & a_{jk} &= x_{jk} - Dx_{jk} = f_{jk} - f_{\bar{j},\bar{k}}. \end{aligned}$$

Their dimensions are $d_+ = \nu(\nu + 1)/2$ and $d_- = \lfloor \nu^2/2 \rfloor$, respectively.

Proof. Theorem 6.2 shows that the covariants span the q -dimensional quotient and that duality acts trivially there. Counting the duality orbits in $\mathcal{V}_n$ therefore gives the stated dimensions. The negative vectors have disjoint antisymmetric syzygy coordinates. For the positive vectors, the strictly lower-triangular pairs likewise have distinct syzygy coordinates. The only repeated diagonal coordinates are those of y_{jj} and $y_{\bar{j},\bar{j}}$, whose difference is $t_{\bar{j},\bar{j}} - t_{jj}$. For $1 \leq j \leq \lfloor \nu/2 \rfloor$, these differences have distinct highest covariant indices $n - j - 2 > j$ with nonzero coefficients, by (32). Both families are consequently independent, and their sizes prove the claim. $\square$

For $1 \leq k \leq j \leq \nu$, let $\mathcal{E}_{jk}(p)$ be the column obtained by retaining only the E -atoms in the linear reconstruction of Appendix A. Precisely, omit every W_n and $H_{n,r}$ term in (38)–(43), but retain the cohomology assembly, Jordan reconstruction, Euler induction, and covariant tail subtraction. The resulting finite formula is recorded in (44).

Lemma 6.4 (Boundary transfer). *For $1 \leq k \leq j \leq \nu$, we have*

$$(33) \quad A_n(p)x_{jk} = \mathcal{E}_{jk}(p).$$

This equality includes every covariant output coordinate.

Proof. Since $x_{jk} = f_{jk} - t_{jk}$, the required identity is equivalent to

$$A_n(p)f_{jk} = A_n(p)t_{jk} + \mathcal{E}_{jk}(p).$$

Every operation in the column reconstruction (38)–(43) is linear in the W_n , $H_{n,s}$, and $E_{j;q,e}$ inputs. Let $\mathcal{G}_{jk}(p)$ be the column obtained by omitting the E -inputs and retaining the W_n and $H_{n,s}$ inputs. By definition of $\mathcal{E}_{jk}$,

$$A_n(p)f_{jk} = \mathcal{G}_{jk}(p) + \mathcal{E}_{jk}(p).$$

We will show that $\mathcal{G}_{jk}(p) = A_n(p)t_{jk}$. Both columns include the covariant outputs obtained after subtracting the syzygy characters.

Matching the weight and fusion inputs. The Borel numerator of the syzygy source is

$$\mathcal{K}_{jk}(z) = \sum_{b=k+1}^n (-1)^{b-k-1} \binom{n}{b} z^{j-b}.$$

Write $\mathcal{W}_p(P)$ for the bounded-weight contribution to the induced character obtained from a numerator $P(z)$ by the projection, Jordan reconstruction, and Euler induction of §3.4. Put $w = z^p$. The reflection identities are

$$\mathcal{W}_p(z^{-a-2}) = -\mathcal{W}_p(z^a), \quad \mathcal{W}_p(z^{-1}) = 0 \quad (a \geq 0).$$

To see the first identity, use the symmetry of $\text{ch } Q$ to note that $(1 - z^2)(z^a + z^{-a-2}) \text{ch } Q$ changes sign under $z \mapsto z^{-1}$. Projection to weights divisible by p preserves this property. After division by $1 - w^2$, the coefficients at w^l and w^{-l-2} agree. These paired terms cancel under Euler induction, which sends them to opposite characters; weight -1 contributes zero. The same argument applied to $(1 - z^2)z^{-1} \text{ch } Q$ proves the second identity.

In $\mathcal{K}_{jk}$, the term with $b = j + 1$ therefore contributes zero. Reflecting the terms with $b \geq j + 2$ and retaining those with $b \leq j$ gives

$$\mathcal{W}_p(\mathcal{K}_{jk}) = \sum_{a=k+1}^{n-1} (-1)^{a-k-1} \binom{n}{b_a} \mathcal{W}_p(z^{c_a}),$$

with exactly the indices (c_a, b_a) defining t_{jk} in (32). Likewise, reindexing (37) gives

$$B_{j,k} = \sum_{a=k+1}^{n-1} (-1)^{a-k-1} \binom{n}{b_a} H_{n,c_a}.$$

After omitting the E -terms, equations (38)–(39) give normalized Tate rows $B_{j,k}$ and $wB_{j,k}$, and syzygy multiplicity $B_{j,k}$ on every diagonal target, with zero on the other syzygy targets. Thus both kinds of input have the alternating coefficients of the covariant tail.

Identifying these inputs with actual covariant columns. We must also justify that Euler induction computes the actual covariant character, rather than a difference involving higher induction. Fix $0 \leq s \leq \nu \leq p - 1$ and put $L_s = (s\omega) \otimes S$. Tensor identity and Kempf vanishing give

$$R \text{Ind}_B^G L_s = T(s) \otimes S.$$

Twisting [10, Lemma 10.2 and Proposition 10.3] by $s\omega$ selects fusion labels s and $p - 2 - s$ in even and odd Tate degree. Their multiplicities agree by Lemma 4.2, giving normalized Tate numerators $H_{n,s}$ and $wH_{n,s}$ in degrees zero and one. The corresponding normalized weights are dominant, so their first higher inductions vanish. For $s = p - 1$, both Tate terms vanish.

In the remainder of this paragraph, write $\mathbf{l} = \text{Ind}_{B(1)}^{G(1)}$. The ordinary comparison of [10, Proposition B.5] identifies $(T(s) \otimes S)^{G_1}$ with $\mathbf{l}(L_s^{B_1})$. Since $G/B = \mathbb{P}^1$ and hence $R^2\mathbf{l} = 0$, its degree-one exact sequence is

$$0 \longrightarrow R^1\mathbf{l}(L_s^{B_1}) \longrightarrow H^1(G_1, T(s) \otimes S) \longrightarrow \mathbf{l}H^1(B_1, L_s) \longrightarrow 0.$$

The Tate comparison in the same proposition, together with $R^1\mathbf{l}\hat{H}^0(B_1, L_s) = 0$, gives

$$\hat{H}^1(G_1, T(s) \otimes S) \cong \mathbf{l}\hat{H}^1(B_1, L_s).$$

Ordinary and Tate cohomology agree in degree one. The middle and right-hand terms of the exact sequence therefore have equal dimensions in each finite internal degree, forcing $R^1\mathbf{l}(L_s^{B_1}) = 0$. Consequently Euler induction computes the actual covariant character from numerator z^s and Tate numerators $H_{n,s}$ and $wH_{n,s}$.

Assembling the columns. For each covariant source $S\{s\}$, the diagonal syzygy multiplicities are $H_{n,s}$. Subtracting their resolution characters from the induced character is the same linear operation

as the subtraction in (43). The matching weight and fusion inputs above therefore give, in every output coordinate,

$$\mathcal{G}_{jk}(p) = \sum_{a=k+1}^{n-1} (-1)^{a-k-1} \binom{n}{b_a} A_n(p) e_{c_a} = A_n(p) t_{jk}.$$

Subtracting this identity from $A_n(p) f_{jk} = \mathcal{G}_{jk}(p) + \mathcal{E}_{jk}(p)$ proves the lemma. The same comparison and Lemma 4.3 include the endpoint characteristic. $\square$

For a vector in either residual space, write $[\cdot]_{\mathcal{B}^\pm}$ for its coordinates in the displayed basis. Define matrices by their source columns:

$$(34) \quad (H_n^+)_{\bullet, (j,k)} = \left[\frac{\mathcal{E}_{jk} + D\mathcal{E}_{jk}}{2} \right]_{\mathcal{B}^+} \quad \text{and} \quad (H_n^-)_{\bullet, (j,k)} = [\mathcal{E}_{jk} - D\mathcal{E}_{jk}]_{\mathcal{B}^-}.$$

These are fixed rational coordinate operations. If $P^\pm$ is the matrix of the corresponding basis in $\mathcal{V}_n$, its left inverse $((P^\pm)^t P^\pm)^{-1} (P^\pm)^t$ extracts the coordinates. For $H_n^\pm$ the covariant coordinates must be retained, since the paired diagonal basis vectors share the same syzygy coordinates.

Theorem 6.5. *For every $n \geq 4$ and prime $p \geq \max\{n-2, 3\}$, we have*

$$(35) \quad \det(tI - A_n(p)) = \prod_{i=0}^{\lfloor n/2 \rfloor - 1} (t - p^{2n-3-2i}) P_n^+(t, p) P_n^-(t, p), \quad \text{with} \quad P_n^\pm(t, p) = \det(tI - H_n^\pm(p)).$$

The matrices $H_n^\pm(p)$ have sizes d_+, d_- and entries in $\mathbb{Q}[p]$ of degree at most $n-1$. Every entry is obtained from the n -variable counts $C_{n,p}$ by fixed rational linear operations with ranges depending only on n . Empty determinants are one.

Proof. Boundary transfer and duality identify (34) with the restrictions of $A_n(p)$ to $\mathcal{N}_n^\pm$. The quotient has the Frobenius weights of Theorem 6.2. A basis adapted to $\mathcal{N}_n$ makes $A_n(p)$ block triangular, proving (35). Each atom $E_{j;s,e} = C_{n,p}((s+e)p - j - 2e)$ is a polynomial of degree at most $n-1$ by Lemma 4.1. Every subsequent operation in (44) and (34) is independent of p , proving the degree bound. $\square$

Under the map that sums coordinates over duality orbits, $\mathcal{N}_n^+$ identifies with the kernel of the matrix W above. Thus H_n^+ and the matrix H of (29) are related by a constant rational change of basis. The boundary construction computes its characteristic factor directly, and also supplies the negative factor absent from (31). The extension data represented by b in (29) is still needed for the signature equations.

For fixed n , every residual eigenvalue is $O_n(p^{n-1})$, since this bounds the norms of $H_n^\pm(p)$. Hence each geometric weight with exponent greater than $n-1$ is a simple root of the full characteristic polynomial for all sufficiently large primes. These are all q geometric weights for odd n and the largest $q-1$ for even n . For $n=4$, the construction gives $H_4^+ = (p(p^2+2)/3)$ and an empty negative block, recovering the spectrum in §3.5.

The boundary counts also have a fiber interpretation. A Koszul calculation gives

$$\mathrm{Tor}_i^S(Q, S_+) \cong k[x_1, \dots, x_n] / (x_1^p, \dots, x_n^p) \otimes \bigwedge^i \langle y_1, \dots, y_n \rangle^{[1]},$$

with exterior generators in internal degree p . These are the Tor modules of the derived intersection of the Frobenius fiber with the incidence fiber underlying the resolutions. Their weight counts enter the boundary reconstruction. No equivariant formality is required.

7. EXPLICIT SIGNATURES

The signature of C_4 was derived by hand in Theorem 3.2. For $n = 5, 6$ the same method gives the following.

Theorem 7.1. *For every prime $p \geq 3$,*

$$s_p(C_5) = \frac{N_5(p)}{168 D_5(p)},$$

where

$$N_5(p) = 1552p^9 + 1552p^8 + 3104p^7 + 3677p^6 + 3947p^5 \\ + 3103p^4 + 2968p^3 + 1551p^2 + 843p + 573,$$

$$D_5(p) = 12p^9 + 12p^8 + 24p^7 + 29p^6 + 31p^5 + 25p^4 + 24p^3 + 13p^2 + 7p + 5.$$

In particular $s_3(C_5) = 1226365/1595624$ and $\lim_{p \rightarrow \infty} s_p(C_5) = 97/126$.

Theorem 7.2. *For every prime $p \geq 5$,*

$$s_p(C_6) = \frac{N_6(p)}{1512 D_6(p)}, \quad N_6(p) = \sum_{j=0}^{30} n_j p^j, \quad D_6(p) = \sum_{j=0}^{30} d_j p^j,$$

with the coefficients of Table 1. Moreover

$$s_p(C_6) = \frac{15067}{22680} - \frac{899}{48600 p^4} + O(p^{-5}).$$

j	n_j	d_j	j	n_j	d_j
0	-2203200	-2496	16	-79081424	-63896
1	-8786880	-9552	17	187168006	203366
2	-21552264	-23400	18	421151974	437262
3	-46964772	-50148	19	615200326	628962
4	-89401548	-94908	20	738355270	749550
5	-152948832	-160684	21	779041292	786528
6	-241372020	-251996	22	761408670	766134
7	-344233233	-356673	23	667306968	669240
8	-470178765	-484345	24	561043200	561600
9	-578750905	-592425	25	409481712	408960
10	-687982955	-700235	26	297903276	297180
11	-731863346	-740450	27	173571840	172800
12	-746891291	-750875	28	108482400	108000
13	-659348687	-657735	29	43392960	43200
14	-531128519	-523915	30	21696480	21600
15	-313172387	-301443			

TABLE 1. The coefficients of N_6 and D_6 in Theorem 7.2.

Proofs of Theorems 7.1 and 7.2. We use the boundary matrices of Section 6.1 to reduce the calculations to three and six unknowns, respectively. Write $d = 2n - 3$, $q = \lfloor n/2 \rfloor$, and let Q_n be the matrix whose columns are the vectors of $\mathcal{B}^+$, ordered by increasing j and then increasing k . Choose the following vectors in the covariant span:

$$u_5 = \frac{5}{6}e_0 + \frac{1}{12}e_1 \quad \text{and} \quad u_6 = \frac{101}{144}e_0 + \frac{5}{36}e_1 + \frac{1}{144}e_2.$$

Formula (25) gives $w_0(u_n) = 1$ and $w_i(u_n) = 0$ for $1 \leq i < q$. Thus the normalized Perron vector has the form $v = u_n + Q_n z$. Set

$$b_n = [A_n(p)u_n - p^d u_n]_{\mathcal{B}^+}, \quad c_n = (Q_n)_{0,\bullet}, \quad \text{and} \quad M_n = p^d I - H_n^+(p).$$

The Riemann–Roch identities and duality place the vector defining b_n in $\mathcal{N}_n^+$. Boundary transfer gives $A_n(p)Q_n = Q_n H_n^+(p)$, so the eigenvector equation becomes

$$M_n z = b_n \quad \text{and} \quad s_p(C_n) = (u_n)_0 + c_n z.$$

The Perron root is simple, hence M_n is invertible at every prime in the stated range.

The accompanying program constructs H_n^+ directly from the E -counts in (44) and (34). To obtain b_n , it uses only the first q covariant-source columns from (6)–(8). It does not assemble the full Frobenius matrix or the negative residual block. The resulting exact linear algebra certifies the polynomial identities $\det M_n = g_n D_n$ and

$$\kappa_n((u_n)_0 \det M_n + c_n \operatorname{adj}(M_n) b_n) = g_n N_n,$$

where $\kappa_5 = 168$, $\kappa_6 = 1512$, and

$$g_5 = \frac{p^6(p-1)^3(p+1)(p^2+p+1)}{12}, \quad g_6 = \frac{p^{14}(p-1)^5(p+1)^3(p^2+1)}{21600}.$$

Here N_n, D_n are exactly the polynomials in the statements. Cramer’s rule now gives $s_p(C_n) = N_n/(\kappa_n D_n)$. Since g_n and $\det M_n$ are nonzero at every prime in range, these fractions are valid throughout that range. Expanding the verified fraction for $n = 6$ at $p = \infty$ gives the stated asymptotic. All calculations use exact polynomial arithmetic; no approximation or interpolation is used. $\square$

Example 7.3 (The three-variable system for C_5). In the order (y_{11}, y_{21}, y_{22}) , the residual matrix is

$$H_5^+(p) = \frac{p}{24} \begin{pmatrix} -p^3 + 14p^2 + p + 10 & -(p-1)(p^2 - 9p + 2) & -(p^2 - 1)(p + 2) \\ 2p(p^2 - 1) & 2p(p^2 + 11) & 2p(p^2 - 1) \\ (p-3)(p-1)(13p+10) & (p-1)(13p^2 - 17p - 6) & (p+1)(13p^2 - 7p + 6) \end{pmatrix}.$$

The right-hand side and free-coordinate row are

$$b_5 = \frac{p(p^2 - 1)}{2520} \begin{pmatrix} -(2p^4 + 2p^2 - 5) \\ 0 \\ 3(12p^4 + 12p^2 + 5) \end{pmatrix}, \quad c_5 = (-10, -5, -5).$$

Thus solving $(p^7 I - H_5^+)z = b_5$ gives $s_p(C_5) = 5/6 - 10z_1 - 5z_2 - 5z_3$. The analogous six-variable system for C_6 , together with polynomial certificates for both systems, is recorded in the ancillary package.

As independent consistency checks, the full-matrix program solves the full normalized system over $\mathbb{Q}$ at $p = 3, 5, 7$ for C_4, C_5 and at $p = 5, 7$ for C_6 , and verifies the Riemann–Roch eigenrow identities for $4 \leq n \leq 8$. These checks merely supplement the symbolic identities. They are not used to interpolate the formulas.

8. CONCLUDING REMARKS

For $p \geq \max\{n - 2, 3\}$, the matrix $A_n(p)$ computes every Frobenius multiplicity of C_n by iteration, and hence the asymptotic proportion of every summand. The characteristic bound is that of [11], so smaller characteristics require further arguments. Theorem 6.2 reduces the signature to a linear system in $\binom{n-2}{2}$ unknowns, but a direct scalar formula for $s_p(C_n)$ for general n remains open. The boundary construction in Theorem 6.5 determines the complete characteristic

polynomial, but full diagonalizability and an explicit description of all eigenspaces remain separate questions.

In the sequel we apply these results to the generic determinantal rings $k[X_{a \times b}]/I_3(X)$. They are weight-zero parts of the rings C_{a+b} for a one-dimensional torus, and retaining that torus character through the calculations of this paper descends them.

The reduced-signature program constructs the systems used in Section 7 from the boundary counts and the first q covariant-source columns. The ancillary package includes their exact matrices, right-hand sides, rational solutions, and cleared polynomial certificates, together with a short certificate checker. The full-matrix program implements the finite assembly formulas of Section 3 and Appendix A as an independent check. It also checks the counting functions W_n and $H_{n,q}$ against direct convolution and fusion multiplication for $4 \leq n \leq 10$, checks the first free multiplicity (21) and Corollary 5.3 for $4 \leq n \leq 10$, and verifies Theorem 6.2 for $4 \leq n \leq 8$. Instructions, the exact matrices and the normalized Perron vectors on duality orbits are included in the ancillary package; see `anc/README.txt` for reproduction commands. A separate check implements (44) and (34) literally. For $4 \leq n \leq 7$, exact comparisons with the independently assembled full matrices verify the boundary-transfer identities (33) as polynomial identities, and the entire characteristic polynomial at several primes. These serve merely as implementation checks.

APPENDIX A. THE SYZGY-COLUMN FORMULAS

This appendix records the finite formulas for the columns of a syzygy source $K\{j, k\}$, used in §§3.3–3.4. It suffices to consider $1 \leq k \leq j \leq n-3$, as sources with $k > j$ are reduced by duality (9). Besides W_n and $H_{n,q}$, the formulas use the bounded-composition atom

$$(36) \quad E_{j;q,e} = C_{n,p}((q+e)p - j - 2e), \quad e \in \{0, 1\},$$

which counts the tuples $t \in [0, p-1]^n$ of total degree $(q+e)p - j - 2e$ carrying the surviving class described below.

Two facts organize the formulas. First, the syzygy targets require no map-rank calculation. After fixing a tuple and a torus weight, the spectral sequence [11, (6.17)] has its E_3 page concentrated in one column [11, Lemma 7.8]. Its E_1 dimensions are fusion coefficients, and the full complex has at most one surviving class, in position $-q$, at tuple degree $(q+e)p - j - 2e$ and row parity $r \equiv q+e \pmod{2}$. Alternating dimensions to the left of position $-k$ give the generic term B below, and adding or removing the unique class gives the indicators in (38) and (39). [11, Proposition 7.6] passes from the truncated complex to the Borel-level syzygy, [11, Corollary 8.4] induces to G_1 , [11, Proposition 3.5(6)] identifies the stable syzygy of M_r in cohomological degree ℓ with $K\{r, \ell\}$, and [11, Theorem 4.3] identifies the resulting ordinary-cohomology counts with the syzygy-target multiplicities. Second, the formulas (40)–(43) for the tilt-free part are the expansion of the Jordan-block identity (10), as explained in §3.4.

For $0 \leq m \leq n$ with $m \neq j+1$, put

$$s_m = \begin{cases} -m, & m \leq j, \\ 1-m, & m \geq j+2, \end{cases} \quad \text{and} \quad h_m = \begin{cases} j-m, & m \leq j, \\ m-j-2, & m \geq j+2. \end{cases}$$

The generic term is

$$(37) \quad B_{j,k} = \sum_{\substack{0 \leq m \leq n, \, m \neq j+1 \\ -(n-1) \leq s_m < -k}} (-1)^{-k-1-s_m} \binom{n}{m} H_{n,h_m}.$$

In the weighted setting of the sequel this term depends on the parity of the cohomological row, but here it does not, since $H_{n,p-2-q} = H_{n,q}$. For $\ell = 0, 1$, the Tate numerators are

$$(38) \quad \begin{aligned} \mathcal{T}_{j,k;\ell,r} &= [r = \ell] B_{j,k} + \sum_{e=0}^1 \sum_{q=k+1}^{n-1-e} (-1)^{q-k} [r = \ell][\ell \equiv q + e] E_{j;q,e} \\ &\quad + \sum_{e=0}^1 \sum_{q=k+1}^{n-1-e} [r = q + \ell - k - 1][\ell - k - 1 \equiv e] E_{j;q,e}, \end{aligned}$$

and for $1 \leq \ell \leq n - 3$, the ordinary-cohomology counts are

$$(39) \quad \begin{aligned} \mathcal{O}_{j,k;\ell,r} &= [r = \ell] B_{j,k} + \sum_{e=0}^1 \sum_{q=k+1}^{n-1-e} (-1)^{q-k} [r = \ell][\ell \equiv q + e] E_{j;q,e} \\ &\quad + [\ell \geq k] \sum_{e=0}^1 \sum_{q=0}^{\min(k, n-1-e)} [r = q + \ell - k][\ell - k \equiv e] E_{j;q,e} \\ &\quad + [\ell \leq k] \sum_{e=0}^1 \sum_{q=k+1}^{n-1-e} [r = q + \ell - k - 1][\ell - k - 1 \equiv e] E_{j;q,e}. \end{aligned}$$

Congruences are modulo 2, and brackets denote indicators.

Set $P_{j,k;c} = 0$ for $c < -n$, and otherwise

$$(40) \quad \begin{aligned} P_{j,k;c} &= \sum_{m=0}^n (-1)^m \binom{n}{m} (\mathcal{T}_{j,k;1,c+m} - \mathcal{T}_{j,k;0,c+m}) \\ &\quad + \sum_{m=k+1}^n (-1)^{m-k-1} \binom{n}{m} (W_n(pc - j + m) - W_n(pc - j + m - 2)). \end{aligned}$$

The reduced pre-induction character has coefficients

$$(41) \quad Z_{j,k;c} = \sum_{t=0}^{n-2} P_{j,k;c-2t} + \sum_{m=0}^n (-1)^m \binom{n}{m} \mathcal{T}_{j,k;0,c+m}, \quad \text{with} \quad -(n-1) \leq c \leq n-3.$$

The syzygy-target entries are

$$(42) \quad A_{K\{r,h\},K\{j,k\}} = \mathcal{O}_{j,k;h,r}.$$

For $1 \leq r, h \leq n - 3$ and $h + 1 \leq m \leq n - 1$, put $(\eta_{r,m}, \zeta_{r,m}) = (r - m, m)$ if $m \leq r$, and $(\eta_{r,m}, \zeta_{r,m}) = (m - r - 1, m + 1)$ if $m > r$. For $0 \leq l \leq n - 3$ the covariant-target entries are

$$(43) \quad \begin{aligned} A_{S\{l\},K\{j,k\}} &= Z_{j,k;l} - Z_{j,k;-l-2} \\ &\quad - \sum_{r,h=1}^{n-3} \sum_{m=h+1}^{n-1} (-1)^{m-h-1} [\eta_{r,m} = l] \binom{n}{\zeta_{r,m}} \mathcal{O}_{j,k;h,r}. \end{aligned}$$

Every expression is a finite sum, with ranges independent of p , of the atoms $H_{n,q}$, W_n and $E_{j;q,e}$ at arguments affine in p .

The boundary vector. The construction in §6.1 uses only the E -terms of these formulas. Write $\mathcal{T}^E$ and $\mathcal{O}^E$ for (38) and (39) with the generic term $B_{j,k}$ omitted. For $\ell = 0, 1$, set

$$J_{j,k;\ell}(w) = \sum_r \mathcal{T}_{j,k;\ell,r}^E w^r \quad \text{and} \quad Z_{j,k}^E(w) = (1 - w^{-1})^n \frac{J_{j,k;1}(w) - w^2 J_{j,k;0}(w)}{1 - w^2}.$$

This is the specialization of (40)–(41) retaining only the E -atoms. Division is expanded at zero. All required coefficients are obtained by replacing $(1 - w^2)^{-1}$ with $\sum_{v=0}^{n-2} w^{2v}$. The numerator has

no exponent below $1 - n$, and the requested exponents range from $-(n - 1)$ to $n - 3$. Using the tail classes of (32), the full vector is

$$(44) \quad \mathcal{E}_{jk}(p) = \sum_{a=0}^{n-3} ([w^a]Z_{j,k}^E - [w^{-a-2}]Z_{j,k}^E)e_a + \sum_{r,h=1}^{n-3} \mathcal{O}_{j,k;h,r}^E x_{rh}.$$

The use of x_{rh} , for all output pairs including $r < h$, incorporates exactly the covariant subtraction in (43). Together with (36), this specifies every input to (34) using only bounded-composition counts.

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